
Learning to Shape the Reward for Agentic Reinforcement Learning via Bilevel Optimization

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Abstract

1 Reinforcement learning for agentic systems often relies on trajectory-level rewards,
2 leading to poor credit assignment in multi-turn tasks. A common approach is to
3 introduce turn-level rewards (such as heuristic signals, LLM judges, or learned
4 rewards) to provide dense feedback. However, such rewards can introduce bias,
5 potentially altering the underlying optimization objective and thus changing the op-
6 timal policy. This paper proposes a principled framework, Bilevel Agentic Reward
7 Shaping (BARS), for learning turn-level rewards that improve credit assignment
8 while preserving the optimal policy. We formulate the problem as a bilevel opti-
9 mization problem where the upper level learns a policy-invariant shaping reward,
10 and the lower level optimizes the policy induced by this reward. This formulation
11 explicitly accounts for how reward updates affect the policy (and trajectory distri-
12 bution) induced by the shaping reward. Building on this framework, we propose
13 an efficient algorithm that alternates between reward alignment and policy update,
14 and establish a convergence rate of $O(1/\sqrt{K})$ under standard assumptions. Empir-
15 ical results demonstrate that our method achieves higher accuracy and improved
16 training stability compared to baseline methods.

17 1 Introduction

18 Large Language Models (LLMs) are evolving from passive text generators into autonomous agents
19 capable of reasoning, acting, and interacting with complex environments through multi-turn inter-
20 actions. Recent systems, including search agents [1], web navigators [2], and coding assistants [3],
21 demonstrate that augmenting LLMs with external tools (e.g., search engines, databases, and code
22 interpreters) can extend their capabilities beyond pure text-based reasoning. These agentic systems
23 require making sequential and multi-turn decisions, where each turn involves generating reasoning
24 traces, selecting actions, and incorporating environmental feedback into its subsequent states.

25 Reinforcement learning (RL) has emerged as a key paradigm for improving LLM behavior in agentic
26 settings. Existing agentic RL approaches largely rely on trajectory-level rewards, where a single
27 reward is assigned at the end of the interaction trajectory. This feedback is typically either an outcome-
28 based reward such as task success [1, 4, 5], or a composite reward that aggregates intermediate and
29 outcome signals into a single scalar [6, 7, 8]. However, such trajectory-level rewards provide only
30 sparse supervision, making it difficult for LLM agents to effectively optimize their policies, and
31 often leading to limited performance gains [9, 10]. To address this sparsity issue, recent works
32 have introduced dense turn-level rewards to provide more fine-grained feedback at each turn of the
33 interaction. These rewards are typically derived from handcrafted heuristics [11], LLM-based judges
34 [9], or learned reward models [10]. Despite their effectiveness in improving learning efficiency,
35 they introduce a fundamental challenge: turn-level rewards can bias the optimization objective
36 and potentially alter the optimal policy defined by the original task reward. Specifically, the task

objective is typically defined by the outcome reward, such as task success. In contrast, the turn-level rewards can deviate from this goal by encouraging auxiliary criteria that are not perfectly aligned with task success. For example, handcrafted rewards may encourage specific reasoning formats not directly related to task success, while LLM judges can exhibit internal biases and inconsistency across contexts [10]. Even when turn-level rewards are learned from outcome signals [10], there is no guarantee that the learned reward preserves the same optimal policy as the original outcome reward.

Beyond this bias issue, we identify a more fundamental limitation in existing reward learning approaches. Prior methods, such as PRIME [12] and iStar [10], typically generate trajectories using the current policy and optimize the reward model based on these trajectories using supervised objectives such as cross-entropy (CE) or direct preference optimization (DPO) losses. However, these approaches do not account for the dependency between the reward and the trajectory distribution induced by the policy, and thus implicitly assume that the trajectory distribution is independent of the reward during optimization. In reality, the trajectory distribution is implicitly determined by the reward, since the policy that generates trajectories is optimized with respect to the reward. As a result, existing approaches optimize reward functions under a trajectory distribution that does not capture this implicit dependency, leading to a gradient for reward updates that fails to account for how changes in the reward propagate through the policy and in turn alter the trajectory distribution.

In this paper, we propose a principled framework for learning turn-level shaping rewards that improves credit assignment while preserving the optimal policy of the original task. The framework, named "Bilevel Agentic Reward Shaping" (BARS), consists of two key components. First, we adopt a potential-based parameterization of the shaping reward, which is guaranteed to preserve the optimal policy [13]. Second, we formulate reward learning as a bilevel optimization problem, where the upper level optimizes the shaping reward r_θ , and the lower level learns the policy π_{r_θ} induced by this reward. A key feature of BARS is that it explicitly captures the full dependency of the upper-level objective on the reward parameter θ through both direct and indirect paths. Specifically, the shaping reward r_θ influences the policy π_{r_θ} , which in turn determines the generated trajectories y , and ultimately affects the upper-level loss $L(\theta, \pi_{r_\theta})$. By accounting for this dependency path $r_\theta \rightarrow \pi_{r_\theta} \rightarrow y \rightarrow L(\theta, \pi_{r_\theta})$, BARS computes the full hypergradient (Figure 1), where the indirect term captures how changes in the reward influence the learned policy and, consequently, the data distribution used to evaluate the objective. Prior methods such as PRIME and iStar only consider the direct term. The upper-level objective is compatible with standard supervised losses such as CE and DPO, while the lower-level optimization is agnostic to the choice of RL optimizer, supporting integrations with standard RL algorithms such as PPO [14], GRPO [15], and RLOO [16]. Our key contributions are summarized as follows:

1. Bilevel Reward Shaping Framework. We introduce a principled framework for learning turn-level shaping rewards in agentic reinforcement learning. Unlike prior approaches that either fail to guarantee policy invariance or ignore the dependency between shaping rewards and the induced trajectory distribution, we propose a policy-invariant parameterization based on potential-based shaping and formulate a bilevel optimization problem that explicitly captures the dependency between the shaping reward, the induced policy, and the resulting trajectories.

2. Hypergradient-based Optimization. We develop an efficient algorithm that alternates between reward alignment and policy update within each iteration. Our method computes hypergradients that capture how reward updates propagate through the policy and influence the induced trajectory distribution, resulting in a more accurate and effective optimization objective.

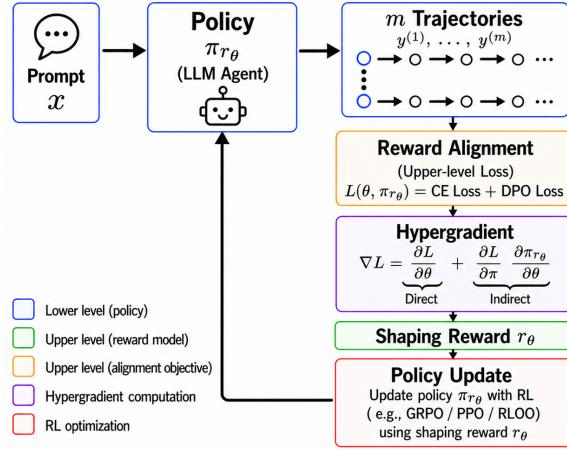


Figure 1: Overview of BARS. The shaping reward r_θ is learned in the upper level, while the policy π_{r_θ} is optimized in the lower level. BARS computes the full hypergradient, which captures both the direct dependency on r_θ and the indirect dependency through the path $r_\theta \rightarrow \pi_{r_\theta} \rightarrow y \rightarrow L(\theta, \pi_{r_\theta})$.

94 **3. Convergence Guarantee and Empirical Gains.** We formally prove that our algorithm converges
 95 at a rate of $O(1/\sqrt{K})$ under standard assumptions. Experimental results demonstrate that our
 96 algorithm achieves higher accuracy and improved training stability compared to the baseline methods.

97 2 Related Work

98 **Agentic RL.** Conventional agentic RL approaches [1, 4, 5] treat multi-turn interactive tasks as
 99 long-horizon single-turn problems and apply standard RL algorithms using outcome rewards such as
 100 task success. However, this trajectory-level formulation often proves inadequate for long-horizon
 101 multi-turn reasoning, as it overlooks the intermediate dynamics of agent-environment interactions,
 102 leading to sparse feedback and poor credit assignment. To address this limitation, subsequent works
 103 introduce process rewards [6, 7, 8, 17, 18, 19] that evaluate properties of intermediate turns, such
 104 as format quality and answer relevance. Nevertheless, these methods still suffer from sparsity
 105 because they aggregate intermediate signals together with the outcome reward into a single scalar
 106 reward delivered only at the end of the interaction trajectory. To provide denser supervision, recent
 107 approaches have shifted towards assigning turn-level rewards to each agent-environment interaction
 108 turn. Such turn-level rewards can be constructed using handcrafted heuristics [11], LLM judges
 109 [9, 20, 21], or learned reward models [10]. However, these turn-level rewards may introduce bias into
 110 the optimization objective and cannot guarantee that the same optimal policy induced by the original
 111 outcome reward is preserved.

112 **Reward Shaping for RL.** Reward shaping improves RL by augmenting the ground-truth reward with
 113 additional shaping signals. Conventional methods rely on rule-based heuristics to design shaping
 114 rewards, including subgoal-based shaping [22, 23], distance-based shaping [24, 25], and potential-
 115 based shaping [13, 26]. Recent learning-based approaches [10, 12, 27] optimize shaping rewards
 116 directly from the ground-truth reward signal. Among them, [10, 12] are most closely related to our
 117 work, as they learn dense shaping rewards from sparse outcome rewards for LLMs. However, these
 118 methods do not account for the implicit dependency between the shaping reward and the generated
 119 trajectory distribution.

120 3 Preliminaries

121 Given a prompt x sampled from a dataset $\mathcal{D}$, an LLM agent interacts with the environment for
 122 T turns. The resulting interaction trajectory is denoted by $y = (y_1, \dots, y_T)$, where each turn
 123 $y_t = (g_t, f_t)$ consists of an LLM response g_t and the corresponding environment feedback f_t . We
 124 model this interaction as a finite-horizon Markov decision process (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{D}, P, \gamma, T, r)$
 125 that includes a state space $\mathcal{S}$, where each state $s_t = (x, y_{<t})$ includes a prompt x and all previous
 126 interactive turns $y_{<t}$. We define the initial state $s_1 = x$ as the prompt sampled from the dataset $\mathcal{D}$.
 127 The action space $\mathcal{A}$ consists of LLM responses, where on each turn the agent selects an action $a_t = g_t$
 128 based on the current state s_t . After the agent generates a response g_t , the environment produces
 129 feedback f_t , and the interaction proceeds to the next state $s_{t+1} = (x, y_{<t}, (g_t, f_t))$. The transition
 130 function $P(s_{t+1}|s_t, a_t)$ captures this interaction process. The discount factor is $\gamma \in [0, 1]$.

131 The reward function r assigns a scalar value to each state-action pair (s_t, a_t) . In many agentic tasks,
 132 the objective is defined by an outcome reward that evaluates the overall success of the interaction.
 133 Such rewards are typically sparse and only provided at the end of the trajectory, i.e., $r(s_t, a_t) = 0$ for
 134 $t < T$. In this paper, we consider a binary success reward as the outcome reward. Specifically, the
 135 success reward r_{succ} assigns a terminal value $r_{\text{succ}}(s_T, a_T) \in \{0, 1\}$, indicating whether the trajectory
 136 successfully solves the task. We discuss how our framework can be adapted to non-binary outcome
 137 rewards in Appendix A.

138 The LLM agent is modeled as a stochastic policy $\pi(a_t|s_t)$, which generates an action a_t given the
 139 current state s_t . The objective is to maximize the expected success rate over prompts sampled from
 140 $\mathcal{D}$. Since the reward is only provided in the terminal turn, this objective reduces to maximizing the
 141 expected terminal success reward over trajectories induced by the policy:

$$\max_{\pi} E_{x \sim \mathcal{D}}^{\pi} \left[r_{\text{succ}}(S_T, A_T) \middle| S_1 = x \right], \quad (1)$$

142 where uppercase (S_t, A_t) denotes random variables and lowercase (s_t, a_t) denotes their realizations.

While this objective (1) precisely captures the task goal, the success reward is sparse, providing supervision only at the end of the trajectory, which makes policy optimization challenging for the LLM agent. To address this issue, recent works introduce turn-level rewards r_{turn} that assign a scalar value at each turn. This leads to the following modified objective:

$$\max_{\pi} E_{x \sim \mathcal{D}} \left[\sum_{t=1}^T \gamma^{t-1} r_{\text{turn}}(S_t, A_t) \mid S_1 = x \right]. \quad (2)$$

While turn-level rewards provide dense supervision and are easier to optimize, they can introduce bias with respect to the original success reward. As a result, the modified objective (2) is not guaranteed to preserve the same optimal policy as the original success-based objective (1).

4 Problem Formulation

In this section, we formulate the problem of learning a turn-level shaping reward that both facilitates policy optimization and preserves the optimal policy of the original success reward. Our formulation consists of two key components: (1) a policy-invariant parameterization of the shaping reward, and (2) a bilevel optimization framework for learning this shaping reward.

4.1 Policy-invariant Parameterization

A common approach to parameterize reward functions for LLMs is to add a linear head on top of the final transformer layer of a language model [28]. However, this generic reward parameterization cannot guarantee that the learned shaping reward function preserves the optimal policy of the original success reward. In contrast, we seek to learn a shaping reward that provably preserves the optimal policy in a principled manner. The policy invariance theorem [13] guarantees that the optimal policy under the success reward r_{succ} remains unchanged if the success reward is shaped as

$$r_{\text{succ}}(s_t, a_t) \rightarrow r_{\text{succ}}(s_t, a_t) + \gamma \Phi(s_{t+1}) - \Phi(s_t),$$

where $\Phi(\cdot)$ is any potential function that depends only on the state. Classic potential-based reward shaping methods [13, 26] construct such potential functions using heuristics or domain knowledge. In contrast, we propose to learn a potential function. Inspired by the recent reward learning approaches [10, 12], we parameterize the potential function as

$$\Phi_{\theta}(s_1) = 0, \quad \Phi_{\theta}(s_{T+1}) = 0, \quad \Phi_{\theta}(s_t) = \gamma^{1-t} \sum_{i=1}^{t-1} \log \frac{\pi_{\theta}(a_i | s_i)}{\pi_{\text{ref}}(a_i | s_i)} \quad 2 \leq t \leq T,$$

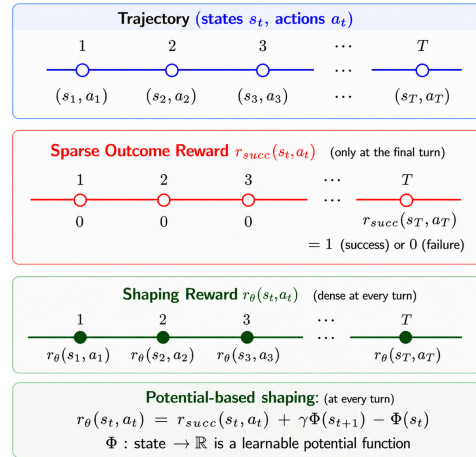
where π_{θ} is a language model with parameter θ , and π_{ref} is a reference policy. Note that the potential function $\Phi_{\theta}(s_t)$ depends only on the state s_t since the trajectory history $\{(s_i, a_i)\}_{i=1}^{t-1}$ is fully contained in s_t . Using this parameterization, the resulting parameterized shaping reward is

$$r_{\theta}(s_t, a_t) = r_{\text{succ}}(s_t, a_t) + \gamma \Phi(s_{t+1}) - \Phi(s_t) = r_{\text{succ}}(s_t, a_t) + \gamma^{1-t} \log \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\text{ref}}(a_t | s_t)}.$$

This potential-based shaping reward r_{θ} (Figure 2) provides turn-level supervision and is guaranteed [13] to preserve the optimal policy of the original success reward r_{succ} .

4.2 Bilevel Optimization

We formulate a bilevel optimization problem where the upper level optimizes shaping reward to align with the success reward and the lower level computes the policy induced by the shaping reward:



- ✓ Provides dense turn-level rewards for better credit assignment.
- ✓ Policy invariant: preserves the optimal policy.

Figure 2: Potential-based reward shaping converts sparse outcome rewards into dense turn-level signals while preserving the optimal policy.

$$\min_{\theta} L(\theta, \pi_{r_{\theta}}) = L_{\text{ce}}(\theta, \pi_{r_{\theta}}) + \lambda L_{\text{dpo}}(\theta, \pi_{r_{\theta}}) \quad (3)$$

$$\text{s.t. } \pi_{r_{\theta}} = \arg \max_{\pi} E_{x \sim \mathcal{D}}^{\pi} \left[\sum_{t=1}^T \gamma^{t-1} \left(r_{\theta}(S_t, A_t) - \beta \log \frac{\pi(A_t|S_t)}{\pi_{\text{ref}}(A_t|S_t)} \right) \middle| S_1 = x \right]. \quad (4)$$

The lower-level problem (4). Given a shaping reward r_{θ} , the lower level solves a regularized RL problem to obtain the corresponding policy $\pi_{r_{\theta}}$, where β is the KL regularization coefficient. Note that $\pi_{r_{\theta}}$ is distinct from the policy π_{θ} used to parameterize the shaping reward. In particular, $\pi_{r_{\theta}}$ has its own model parameters. We use this notation $\pi_{r_{\theta}}$ to highlight the implicit dependency of the lower-level induced policy on the shaping reward r_{θ} .

The upper-level problem (3). To align the shaping reward with the binary success reward, we optimize a combination of a cross-entropy (CE) loss and a DPO-style contrastive loss: $L(\theta, \pi_{r_{\theta}}) = L_{\text{ce}}(\theta, \pi_{r_{\theta}}) + \lambda L_{\text{dpo}}(\theta, \pi_{r_{\theta}})$, where λ is a weighting coefficient. These two losses capture complementary aspects of reward learning: absolute outcome alignment and contrastive discrimination.

The CE loss encourages the shaping reward to reflect the success likelihood of individual trajectories. Given a prompt x , a trajectory y , and the success reward $r_{\text{succ}}(x, y) \in \{0, 1\}$, the CE loss is

$$L_{\text{ce}}(\theta, \pi_{r_{\theta}}) \triangleq -E_{x \sim \mathcal{D}, y \sim \pi_{r_{\theta}}(\cdot|x)} \left[r_{\text{succ}}(x, y) \cdot \log \sigma(r_{\theta}(x, y)) + (1 - r_{\text{succ}}(x, y)) \cdot \log [1 - \sigma(r_{\theta}(x, y))] \right],$$

where $\sigma(\cdot)$ is the sigmoid function, $\pi_{r_{\theta}}(y|x) \triangleq \prod_{t=1}^T \pi_{r_{\theta}}(y_t|x, y_{<t})$ is the probability of policy $\pi_{r_{\theta}}$ generating a trajectory y to the prompt x , $r_{\theta}(x, y) \triangleq \sum_{t=1}^T \gamma^{t-1} r_{\theta}(s_t, a_t)$, and $r_{\text{succ}}(x, y) \triangleq \sum_{t=1}^T \gamma^{t-1} r_{\text{succ}}(s_t, a_t)$. While the CE loss aligns with success likelihood, it provides weak supervision for distinguishing between trajectories. To address this, DPO loss accounts for relative preference, encouraging successful trajectories to receive higher rewards than unsuccessful ones:

$$L_{\text{dpo}}(\theta, \pi_{r_{\theta}}) \triangleq -E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_{r_{\theta}}(\cdot|x)} \left[\log \sigma(r_{\theta}(x, y^+) - r_{\theta}(x, y^-)) \right],$$

where y^+ and y^- denote trajectories with success labels 1 and 0, respectively. When $\pi_{r_{\theta}}$ generates all positive or all negative trajectories, the DPO loss is zero and the upper level relies on the CE loss.

Remark (distinctions from prior works). While PRIME [12] and iStar [10] also use CE or DPO losses for reward learning, our approach differs in two key aspects. First, our potential-based reward parameterization guarantees policy invariance, whereas their parameterizations do not provide such guarantees. Second, our bilevel formulation explicitly accounts for the dependency between the shaping reward and generated trajectories through the hyperparameterized policy $\pi_{r_{\theta}}$. This allows us to compute the full gradient for reward optimization, which captures not only the direct effect of r_{θ} but also its indirect effect through the induced policy (see Section 5 for details).

5 Algorithm and Convergence Guarantee

We develop Algorithm 1 to solve the bilevel optimization problem (3)–(4). At a high level, the algorithm alternates between (i) a policy update step that solves the lower-level problem (4), and (ii) a reward alignment step that solves the upper-level problem (3). A key component of our method is the full hypergradient (i.e., the full gradient of the upper-level objective with respect to θ), which captures both the direct effect of the shaping reward and its indirect effect through the induced policy.

Algorithm 1 Bilevel Agentic Reward Shaping (BARS)

Input: Dataset $\mathcal{D}$, success reward r_{succ} , initial shaping parameter θ_0 , and reference policy π_{ref} .

Output: Learned shaping parameter θ_K and policy π_K .

- 1: Initialize policy $\pi_0 = \pi_{\text{ref}}$.
 - 2: **for** $k = 0, 1, \dots, K - 1$ **do**
 - 3: **Data sample:** Sample a batch of B prompts $\{x^{(n)}\}_{n=1}^B$ from $\mathcal{D}$.
 - 4: **Policy rollout:** For each prompt $x^{(n)}$, generate m trajectories $\{y^{(n,j)}\}_{j=1}^m$ using policy π_k .
 - 5: **Policy update:** Perform one-step policy optimization to obtain π_{k+1} .
 - 6: **Reward alignment:** Compute the hypergradient in Lemmas 1–2 and update shaping parameter via $\theta_{k+1} = \theta_k - \alpha [g_{\text{ce}}(\theta_k) + \lambda g_{\text{dpo}}(\theta_k)]$.
 - 7: **end for**
-

At each iteration k , BARS first samples B prompts from the dataset $\mathcal{D}$ and then uses the current policy π_k to generate m trajectories $\{y^{(n,j)}\}_{j=1}^m$ for each prompt $x^{(n)}$, where each trajectory corresponds to a sequence of state-action pairs $\{(s_t^{(n,j)}, a_t^{(n,j)})\}_{t=1}^T$. These m trajectories are reused for both policy update and reward alignment, enabling efficient training without additional rollouts.

Policy Update. The lower-level problem (4) can be completely solved by soft policy iteration [29]. However, it is computationally expensive to run multiple soft policy iterations to fully solve the lower-level problem at each iteration k . In contrast, we use only a one-step soft policy iteration to partially solve the lower-level problem to obtain an updated policy π_{k+1} . In practice, this one-step soft policy iteration can be instantiated by one-step GRPO/PPO/RLOO [15, 14, 16].

Reward Alignment. To solve the upper-level problem (3), we compute the full hypergradient of $L(\theta, \pi_{r_\theta})$ with respect to θ . Taking the CE loss L_{ce} as an example, we can rewrite it as $L_{\text{ce}}(\theta, \pi_{r_\theta}) = E_{x \sim \mathcal{D}} [\sum_y \pi_{r_\theta}(y|x) \cdot l_{\text{ce}}(\theta; (x, y))]$, where $l_{\text{ce}}(\theta; (x, y)) \triangleq -r_{\text{succ}}(x, y) \cdot \log \sigma(r_\theta(x, y)) - (1 - r_{\text{succ}}(x, y)) \cdot \log[1 - \sigma(r_\theta(x, y))]$ is the individual CE loss of (x, y) . Since the trajectory distribution depends on θ through the induced policy π_{r_θ} , the hypergradient consists of two components:

$$\nabla L_{\text{ce}}(\theta, \pi_{r_\theta}) = \underbrace{E_{x \sim \mathcal{D}} \left[\sum_y \pi_{r_\theta}(y|x) \cdot \nabla_\theta l_{\text{ce}}(\theta; (x, y)) \right]}_{\text{direct term}} + \underbrace{E_{x \sim \mathcal{D}} \left[\sum_y \nabla_\theta \pi_{r_\theta}(y|x) \cdot l_{\text{ce}}(\theta; (x, y)) \right]}_{\text{indirect term}}.$$

The first term captures the direct dependency of the loss on θ , while the second term captures the indirect effect through the trajectory distribution induced by π_{r_θ} . PRIME [12] and iStar [10] only consider the first term, ignoring the implicit dependency of π_{r_θ} on θ . In contrast, we consider the full hypergradient and use the sampled trajectories in the "Policy rollout" step to estimate the hypergradient. Since the upper-level loss in (3) consists of a CE loss and a DPO loss, we provide the hypergradients of these two losses separately.

Lemma 1. *The hypergradient of the CE loss is estimated as*

$$g_{\text{ce}}(\theta) = \frac{1}{Bm} \sum_{n=1}^B \sum_{j=1}^m \left[\nabla_\theta l_{\text{ce}}(\theta; (x^{(n)}, y^{(n,j)})) + l_{\text{ce}}(\theta; (x^{(n)}, y^{(n,j)})) \cdot \Delta_{\text{ce}}^{(n,j)} \right],$$

where $l_{\text{ce}}(\theta; (x, y))$ is the CE loss of (x, y) whose expression and gradient are in Appendix B.1, and $\Delta_{\text{ce}}^{(n,j)}$ is a correction term whose full expression is in Appendix B.1.

For the DPO loss, we form preference pairs. Let $\bar{m}^{(n)}$ denote the number of successful trajectories among the m samples for the prompt $x^{(n)}$. Then, we construct $\bar{m}^{(n)}(m - \bar{m}^{(n)})$ preference pairs $\{(y^{(n,\bar{j},+)}, y^{(n,\bar{j},-)})\}_{\bar{j}=1}^{\bar{m}^{(n)}(m - \bar{m}^{(n)})}$, where each pair consists of a successful trajectory y^+ and an unsuccessful trajectory y^- . If the m trajectories are all positive or all negative, the DPO loss with respect to this prompt $x^{(n)}$ is 0.

Lemma 2. *The hypergradient of the DPO loss is estimated as*

$$g_{\text{dpo}}(\theta) = \frac{1}{B} \sum_{n=1}^B \frac{1}{\bar{m}^{(n)}(m - \bar{m}^{(n)})} \sum_{\bar{j}=1}^{\bar{m}^{(n)}(m - \bar{m}^{(n)})} \left[\nabla_\theta l_{\text{dpo}}(\theta; (x^{(n)}, y^{(n,\bar{j},+)}, y^{(n,\bar{j},-)})) + l_{\text{dpo}}(\theta; (x^{(n)}, y^{(n,\bar{j},+)}, y^{(n,\bar{j},-)})) \cdot \Delta_{\text{dpo}}^{(n,\bar{j})} \right],$$

where $l_{\text{dpo}}(\theta; (x, y^+, y^-))$ is the DPO loss of (x, y^+, y^-) whose expression and gradient are in Appendix B.2, and $\Delta_{\text{dpo}}^{(n,\bar{j})}$ is a correction term whose full expression is in Appendix B.2.

The correction terms in Lemma 1 and Lemma 2 account for how changes in the shaping reward affect the trajectory distribution. We then update the shaping parameter via gradient descent: $\theta_{k+1} = \theta_k - \alpha [g_{\text{ce}}(\theta_k) + \lambda g_{\text{dpo}}(\theta_k)]$, where α is a constant step size.

We now provide our convergence guarantee under the following assumption.

Assumption 1. *The policy satisfies $\|\nabla_\theta \log \pi_\theta(a|s)\| \leq C$ and $\|\nabla_{\theta\theta}^2 \log \pi_\theta(a|s)\| \leq \bar{C}$, and the reward satisfies $|r_\theta(s, a)| \leq \tilde{C}$ for all (s, a) , where C , $\bar{C}$, and $\tilde{C}$ are positive constants.*

Assumption 1 is a standard assumption in RL [30, 31], and we quantify the convergence below:

Theorem 1. Suppose Assumption 1 holds and the step size is $\alpha \propto 1/\sqrt{K}$, then it holds that

$$\frac{1}{K} \sum_{k=0}^{K-1} |\log \pi_{k+1}(a|s) - \log \pi_{r_{\theta_k}}(a|s)| \leq O\left(\frac{1}{\sqrt{K}}\right) + O\left(\frac{1}{K}\right) = O\left(\frac{1}{\sqrt{K}}\right), \quad \forall (s, a) \in \mathcal{S} \times \mathcal{A}$$

$$\frac{1}{K} \sum_{k=0}^{K-1} E\left[\|\nabla L(\theta_k, \pi_k)\|^2\right] \leq O\left(\frac{1}{\sqrt{K}}\right) + O\left(\frac{1}{K}\right) = O\left(\frac{1}{\sqrt{K}}\right).$$

Theorem 1 shows that (i) the iteratively updated policy π_k tracks the optimal lower-level solution $\pi_{r_{\theta_k}}$, despite only performing one-step policy updates; and (ii) the shaping parameter θ_k converges to the set of stationary points of the upper-level objective. The proof is in Appendix B.

6 Experiments

In this section, we demonstrate that BARS achieves higher accuracy and improved training stability. Training details are included in Appendix C.

Models. We consider two base models of different scales: Qwen2.5-3B and Llama3.1-8B.

Benchmarks. We evaluate BARS on two search benchmarks, NQ-Search and NQ-HotpotQA [1]. NQ-Search serves as the in-domain benchmark, where both the training and test sets are drawn from NQ [32]. NQ-HotpotQA evaluates generalization to out-of-domain, multi-step reasoning settings: its training set is constructed from NQ and HotpotQA [33], while its test set spans multiple sources, including general question answering (NQ, TriviaQA [34], and PopQA [35]), and multi-hop question answering (HotpotQA and 2WikiMultiHopQA [36]). For both search benchmarks, the outcome reward is a binary “exact match” signal indicating whether the generated answer exactly matches the ground-truth answer. The number of turns is 2 for NQ-Search and 4 for NQ-HotpotQA, since HotpotQA and 2WikiMultiHopQA involve multi-hop questions.

RL Optimizers. Our method is compatible with standard RL optimizers. In the main experiments, we use GRPO [15] as the default optimizer. To validate optimizer-agnostic compatibility, we further evaluate our method with PPO [14] and RLOO [16].

Baselines. We compare against four representative baselines: (1) **RL+outcome reward (OR)** maximizes the binary “exact match” reward; (2) **RL+turn-level LLM judge (LLMJ)** [9] uses OpenAI’s GPT-OSS-120B [37] as a judge to provide turn-level rewards based on format, information quality, and search efficiency (the judge prompt is provided in Appendix C.1); (3) **PRIME** [12] learns a shaping reward using the CE loss (although it is originally designed for RLOO, we adapt it to support GRPO and PPO); and (4) **iStar** [10] learns a shaping reward using the DPO loss.

Evaluation. We report success rate, i.e., the proportion of prompts for which the model’s predicted answer exactly matches the ground truth. All results are computed from three random seeds.

Table 1: Performance comparison (success rate) on NQ-Search and NQ-HotpotQA benchmarks. † denotes best results before training collapse. All scores are reported as percentages.

Model	Method	NQ-Search	NQ-HotpotQA				
			NQ	HotpotQA	TriviaQA	PopQA	2WikiMultiHopQA
Qwen2.5-3B	OR	40.9 ± 0.3	43.0 ± 0.2†	33.0 ± 0.4†	59.3 ± 0.3†	40.9 ± 0.5†	27.5 ± 0.2†
	LLMJ	41.9 ± 0.5†	36.2 ± 0.4†	30.7 ± 0.4†	54.5 ± 0.5†	38.5 ± 0.6†	29.9 ± 0.4†
	PRIME	45.3 ± 0.3†	46.6 ± 0.2†	34.3 ± 0.2†	59.2 ± 0.3†	44.7 ± 0.2†	29.2 ± 0.3†
	iStar	44.0 ± 0.2	43.3 ± 0.2	32.1 ± 0.3	58.1 ± 0.2	42.7 ± 0.3	26.1 ± 0.4
	BARS	47.8 ± 0.3	46.9 ± 0.3	42.1 ± 0.2	62.2 ± 0.2	46.8 ± 0.3	38.6 ± 0.2
Llama3.1-8B	OR	0.1 ± 0.0†	0.1 ± 0.0†	0.1 ± 0.0†	0.1 ± 0.0†	0.1 ± 0.0†	0.1 ± 0.0†
	LLMJ	38.7 ± 0.5†	32.3 ± 0.3†	27.8 ± 0.5†	59.0 ± 0.4†	27.1 ± 0.3†	27.9 ± 0.4†
	PRIME	44.6 ± 0.2	44.5 ± 0.6	31.7 ± 0.3	60.6 ± 0.4	43.5 ± 0.3	28.8 ± 0.3
	iStar	45.9 ± 0.3†	42.2 ± 0.5†	32.4 ± 0.4†	59.3 ± 0.2†	43.9 ± 0.4†	27.6 ± 0.3†
	BARS	48.7 ± 0.2	46.8 ± 0.2	36.6 ± 0.1	62.8 ± 0.2	46.4 ± 0.2	30.8 ± 0.2

6.1 Main Results

The results in Table 1 demonstrate that BARS achieves the best performance across both models and benchmarks. Taking the NQ-Search benchmark as an example, on Qwen2.5-3B, BARS improves over OR by 6.9% and surpasses the strongest baseline PRIME by 2.5%. On Llama3.1-8B, BARS further reaches 48.7%, achieving the best overall success rate. In contrast, OR performs poorly, most notably for Llama3.1-8B, where training completely fails. This highlights the difficulty of learning from sparse binary supervision without intermediate credit assignment. While LLM-based turn-level rewards (LLMJ) provide denser feedback, they yield only marginal improvement because they suffer from training instability due to the internal bias and inconsistency of the LLM judge. Methods that learn reward models (PRIME, iStar, and BARS) generally perform better, underscoring the importance of learned shaping rewards. Among these reward learning methods, BARS consistently achieves the strongest performance across models and benchmarks.

6.2 Training Dynamics

Figure 3 further illustrates the training dynamics of BARS and the baselines. On both training and test curves, BARS consistently maintains stable performance throughout training, avoiding the large fluctuations observed in several baselines. In contrast, OR improves slowly due to sparse outcome-level supervision. LLMJ exhibits significant instability, with large fluctuations and even severe performance drops, suggesting that LLM-based turn-level rewards can be noisy and inconsistent. While PRIME and iStar both improve upon OR, their optimization dynamics differ: PRIME shows noticeable instability during training, whereas iStar exhibits relatively smoother early-stage behavior but plateaus at a lower performance than BARS after around 100 training steps. We include the training and test curves on the NQ-HotpotQA benchmark in Appendix C.2. Across benchmarks, BARS consistently demonstrates more stable training dynamics compared to the baselines. These results indicate that BARS provides a more reliable optimization signal, leading to stable and consistent policy improvement.



Figure 3: Training and test curves on Qwen2.5-3B and NQ-Search.

6.3 Compatibility with RL Optimizers

Table 2: Performance comparison on Qwen2.5-3B under different RL optimizers.

RL optimizer	Method	NQ-Search	NQ-HotpotQA				
			NQ	HotpotQA	TriviaQA	PopQA	2WikiMultiHopQA
PPO	OR	42.4 ± 0.2	40.1 ± 0.3	31.5 ± 0.2	59.0 ± 0.3	39.7 ± 0.2	27.2 ± 0.4
	LLMJ	43.0 ± 0.3	40.8 ± 0.3	32.2 ± 0.2	59.6 ± 0.2	40.9 ± 0.2	28.3 ± 0.3
	PRIME	45.4 ± 0.3	41.6 ± 0.2	29.2 ± 0.4	59.9 ± 0.2	39.8 ± 0.3	29.5 ± 0.2
	iStar	45.7 ± 0.5	41.4 ± 0.3	32.6 ± 0.2	59.8 ± 0.3	40.7 ± 0.4	31.1 ± 0.3
	BARS	48.1 ± 0.2	43.0 ± 0.2	35.6 ± 0.3	61.1 ± 0.2	43.9 ± 0.3	33.1 ± 0.3
RLOO	OR	38.2 ± 0.3†	40.4 ± 0.2†	32.2 ± 0.3†	57.3 ± 0.2†	40.1 ± 0.3†	26.9 ± 0.3†
	LLMJ	39.6 ± 0.6†	39.2 ± 0.2†	33.7 ± 0.3†	55.3 ± 0.2†	31.9 ± 0.2†	24.7 ± 0.2†
	PRIME	44.9 ± 0.2†	44.2 ± 0.3†	33.3 ± 0.2†	60.1 ± 0.3†	42.6 ± 0.4†	26.7 ± 0.2†
	iStar	44.7 ± 0.1	43.8 ± 0.2	33.5 ± 0.3	57.6 ± 0.3	41.7 ± 0.1	27.2 ± 0.4
	BARS	47.5 ± 0.3	46.0 ± 0.3	35.7 ± 0.1	62.7 ± 0.3	46.1 ± 0.4	29.2 ± 0.2

While Table 1 reports results using GRPO, BARS is agnostic to the choice of RL optimizers and remains effective across different optimizers. Table 2 shows that BARS achieves the best performance under both PPO and RLOO. For example, under RLOO, BARS improves over the strongest baseline PRIME by 2.6% on NQ-Search and yields consistent gains across NQ-HotpotQA subsets. Similar trends hold under PPO, where BARS outperforms the baselines.

6.4 Ablation Studies

We conduct ablation studies to understand the contribution of two key components in BARS: (i) the complete hypergradient, and (ii) the potential-based reward parameterization. To evaluate the role of the hypergradient, we compare BARS with a variant that removes the indirect term and optimizes the reward using only the direct gradient. To study the impact of reward parameterization, we replace the potential-based shaping reward with a standard reward model implemented as a linear head on top of the final transformer layer of π_{ref} [28]. The results in Table 3 show that removing either component consistently degrades performance across models and benchmarks. In particular, using only the direct gradient leads to noticeable drops, confirming the importance of accounting for the dependency between the reward and the induced policy. Replacing the potential-based parameterization with a linear reward model results in even larger performance degradation in most cases, suggesting the importance of preserving policy invariance via the potential-based reward shaping.

Table 3: Ablation studies (with GRPO as the RL optimizer). The reported scores are means.

Model	Ablation	NQ-Search	NQ-HotpotQA				
			NQ	HotpotQA	TriviaQA	PopQA	2WikiMultiHopQA
Qwen2.5-3B	direct gradient	46.1(↓ 1.7)	46.2(↓ 0.7)	37.2(↓ 4.9)	59.8(↓ 2.4)	43.8(↓ 3.0)	30.4(↓ 8.2)
	linear parameterization	45.8(↓ 2.0)	45.8(↓ 1.1)	36.9(↓ 5.2)	60.1(↓ 2.1)	40.7(↓ 6.1)	27.9(↓ 10.7)
Llama3.1-8B	direct gradient	46.2(↓ 2.5)	45.0(↓ 1.8)	34.4(↓ 2.2)	61.3(↓ 1.5)	44.7(↓ 1.7)	29.4(↓ 1.4)
	linear parameterization	45.6(↓ 3.1)	44.2(↓ 2.6)	32.7(↓ 3.9)	58.7(↓ 4.1)	43.0(↓ 3.4)	27.7(↓ 3.1)

6.5 Sensitivity Analyses

We conduct sensitivity analyses to evaluate BARS with respect to key hyperparameters, including (i) the number of rollouts m , (ii) the KL regularization coefficient β , and (iii) the weighting coefficient λ in the upper-level objective. We focus on the Qwen2.5-3B model on the NQ-Search benchmark. The results in Table 4 suggest the following. (i) increasing the number of rollouts improves the performance as it provides more diverse trajectories for reward learning and policy optimization. However, the gains gradually plateau after $m = 8$. (ii) Performance improves as β increases from 0.001 to 0.005, indicating that moderate regularization helps training. However, overly large β values (e.g., 0.015 and 0.020) degrade performance, as strong regularization restricts policy updates. (iii) A moderate weighting coefficient (e.g., $\lambda = 0.05$ or 0.10) achieves the best performance, while only considering CE loss ($\lambda = 0$) or DPO loss leads to large degradation. This indicates that combining the complementary supervision (absolute alignment and contrastive discrimination) is important for reward learning.

Table 4: Sensitivity analysis (with GRPO as the RL optimizer).

Rollout number m	4	6	8	10	12
Success rate	44.9 ± 0.3	46.6 ± 0.2	47.8 ± 0.3	48.2 ± 0.2	48.4 ± 0.3
KL coefficient β	0.001	0.005	0.010	0.015	0.020
Success rate	45.8 ± 0.4	47.8 ± 0.3	47.6 ± 0.2	46.2 ± 0.3	44.5 ± 0.2
Weighting coefficient λ	0	0.05	0.10	0.15	DPO only
Success rate	46.8 ± 0.2	47.5 ± 0.2	47.8 ± 0.3	46.5 ± 0.5	45.8 ± 0.6

7 Conclusion

This paper proposes BARS, a principled framework for learning turn-level shaping rewards in agentic RL that addresses the fundamental challenge of sparse supervision while avoiding bias in the optimization objective. By combining a policy-invariant, potential-based reward parameterization with a bilevel optimization formulation, our approach explicitly captures the dependency between rewards, policies, and the induced trajectory distribution through a full hypergradient. This enables more accurate reward learning compared to prior methods that do not account for this dependency. We theoretically prove that the algorithm converges at a rate of $O(1/\sqrt{K})$ under standard assumptions. Empirical results demonstrate that BARS achieves improved training stability and superior performance across multiple models and benchmarks.

References

- [1] B. Jin, H. Zeng, Z. Yue, J. Yoon, S. O. Arik, D. Wang, H. Zamani, and J. Han, “Search-R1: Training llms to reason and leverage search engines with reinforcement learning,” in *Conference on Language Modeling*, 2025.
- [2] H. Furuta, K.-H. Lee, O. Nachum, Y. Matsuo, A. Faust, S. S. Gu, and I. Gur, “Multimodal web navigation with instruction-finetuned foundation models,” in *International Conference on Learning Representations*, 2024.
- [3] K. Zhang, J. Li, G. Li, X. Shi, and Z. Jin, “Codeagent: Enhancing code generation with tool-integrated agent systems for real-world repo-level coding challenges,” in *Annual Meeting of the Association for Computational Linguistics*, pp. 13643–13658, 2024.
- [4] J. Feng, S. Huang, X. Qu, G. Zhang, Y. Qin, B. Zhong, C. Jiang, J. Chi, and W. Zhong, “Retool: Reinforcement learning for strategic tool use in LLMs,” *arXiv preprint arXiv:2504.11536*, 2025.
- [5] M. Chen, L. Sun, T. Li, C. Zhu, H. Wang, J. Z. Pan, W. Zhang, H. Chen, F. Yang, Z. Zhou, *et al.*, “Learning to reason with search for LLMs via reinforcement learning,” in *Advances in Neural Information Processing Systems*, 2026.
- [6] X. Li, H. Zou, and P. Liu, “ToRL: Scaling tool-integrated RL,” *arXiv preprint arXiv:2503.23383*, 2025.
- [7] H. Wang, C. Qian, W. Zhong, X. Chen, J. Qiu, S. Huang, B. Jin, M. Wang, K.-F. Wong, and H. Ji, “OTC: Optimal tool calls via reinforcement learning,” *arXiv e-prints*, pp. arXiv–2504, 2025.
- [8] Z. Wang, K. Wang, Q. Wang, P. Zhang, L. Li, Z. Yang, X. Jin, K. Yu, M. N. Nguyen, L. Liu, *et al.*, “Ragen: Understanding self-evolution in LLM agents via multi-turn reinforcement learning,” *arXiv preprint arXiv:2504.20073*, 2025.
- [9] Q. Wei, S. Zeng, C. Li, W. Brown, O. Frunza, W. Deng, A. Schneider, Y. Nevmyvaka, Y. K. Zhao, A. Garcia, *et al.*, “Reinforcing multi-turn reasoning in LLM agents via turn-level reward design,” *arXiv preprint arXiv:2505.11821*, 2025.
- [10] X. Liu, K. Wang, Y. Wu, F. Huang, Y. Li, J. Zhang, and J. Jiao, “Agentic reinforcement learning with implicit step rewards,” in *International Conference on Learning Representations*, 2026.
- [11] Z. Zhang, Z. Chen, M. Li, Z. Tu, and X. Li, “RLVMR: Reinforcement learning with verifiable meta-reasoning rewards for robust long-horizon agents,” in *International Conference on Learning Representations*, 2026.
- [12] G. Cui, L. Yuan, Z. Wang, H. Wang, Y. Zhang, J. Chen, W. Li, B. He, Y. Fan, T. Yu, Q. Xu, W. Chen, J. Yuan, H. Chen, K. Zhang, X. Lv, S. Wang, Y. Yao, X. Han, H. Peng, Y. Cheng, Z. Liu, M. Sun, B. Zhou, and N. Ding, “Process reinforcement through implicit rewards,” *arXiv preprint arXiv:2502.01456*, 2025.
- [13] A. Y. Ng, D. Harada, and S. Russell, “Policy invariance under reward transformations: Theory and application to reward shaping,” in *International Conference on Machine Learning*, vol. 99, pp. 278–287, 1999.
- [14] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347*, 2017.
- [15] Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, X. Bi, H. Zhang, M. Zhang, Y. Li, Y. Wu, *et al.*, “Deepseekmath: Pushing the limits of mathematical reasoning in open language models,” *arXiv preprint arXiv:2402.03300*, 2024.
- [16] A. Ahmadian, C. Cremer, M. Gallé, M. Fadaee, J. Kreutzer, O. Pietquin, A. Üstün, and S. Hooker, “Back to basics: Revisiting REINFORCE-style optimization for learning from human feedback in LLMs,” in *Annual Meeting of the Association for Computational Linguistics*, pp. 12248–12267, 2024.

- [17] C. Qian, E. C. Acikgoz, Q. He, H. Wang, X. Chen, D. Hakkani-Tür, G. Tur, and H. Ji, “ToolRL: Reward is all tool learning needs,” *Advances in Neural Information Processing Systems*, 2025.
- [18] J. Singh, R. Magazine, Y. Pandya, and A. Nambi, “Agentic reasoning and tool integration for LLMs via reinforcement learning,” *arXiv preprint arXiv:2505.01441*, 2025.
- [19] S. Zhang, Y. Dong, J. Zhang, J. Kautz, B. Catanzaro, A. Tao, Q. Wu, Z. Yu, and G. Liu, “Nemotron-research-tool-nl: Tool-using language models with reinforced reasoning,” *arXiv preprint arXiv:2505.00024*, vol. 2, no. 10, 2025.
- [20] X. Liu, K. Wang, Y. Li, Y. Wu, W. Ma, A. Kong, F. Huang, J. Jiao, and J. Zhang, “Epo: Explicit policy optimization for strategic reasoning in llms via reinforcement learning,” in *Annual Meeting of the Association for Computational Linguistics*, pp. 15371–15396, 2025.
- [21] K. Zha, Z. Gao, M. Shen, Z.-W. Hong, D. S. Boning, and D. Katabi, “RL tango: Reinforcing generator and verifier together for language reasoning,” *Advances in Neural Information Processing Systems*, 2025.
- [22] S. Singh, R. L. Lewis, and A. G. Barto, “Where do rewards come from,” in *Annual Conference of the Cognitive Science Society*, pp. 2601–2606, 2009.
- [23] G. Konidaris and A. Barto, “Skill discovery in continuous reinforcement learning domains using skill chaining,” *Advances in Neural Information Processing Systems*, vol. 22, 2009.
- [24] M. J. Mataric, “Reward functions for accelerated learning,” in *International Conference on Machine Learning*, pp. 181–189, 1994.
- [25] A. Laud and G. DeJong, “The influence of reward on the speed of reinforcement learning: An analysis of shaping,” in *International Conference on Machine Learning*, pp. 440–447, 2003.
- [26] E. Wiewiora, “Potential-based shaping and q-value initialization are equivalent,” *Journal of Artificial Intelligence Research*, vol. 19, pp. 205–208, 2003.
- [27] S. Liu and M. Zhu, “Utility: Utilizing explainable reinforcement learning to improve reinforcement learning,” in *International Conference on Learning Representations*, 2025.
- [28] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, *et al.*, “Training language models to follow instructions with human feedback,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 27730–27744, 2022.
- [29] T. Haarnoja, H. Tang, P. Abbeel, and S. Levine, “Reinforcement learning with deep energy-based policies,” in *International Conference on Machine Learning*, pp. 1352–1361, 2017.
- [30] K. Zhang, A. Koppel, H. Zhu, and T. Basar, “Global convergence of policy gradient methods to (almost) locally optimal policies,” *SIAM Journal on Control and Optimization*, vol. 58, no. 6, pp. 3586–3612, 2020.
- [31] H. Kumar, A. Koppel, and A. Ribeiro, “On the sample complexity of actor-critic method for reinforcement learning with function approximation,” *Machine Learning*, vol. 112, no. 7, pp. 2433–2467, 2023.
- [32] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, “Dense passage retrieval for open-domain question answering,” in *Conference on Empirical Methods in Natural Language Processing*, pp. 6769–6781, 2020.
- [33] Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. Cohen, R. Salakhutdinov, and C. D. Manning, “HotpotQA: A dataset for diverse, explainable multi-hop question answering,” in *Conference on Empirical Methods in Natural Language Processing*, pp. 2369–2380, 2018.
- [34] M. Joshi, E. Choi, D. S. Weld, and L. Zettlemoyer, “TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension,” in *Annual Meeting of the Association for Computational Linguistics*, pp. 1601–1611, 2017.

- 446 [35] A. Mallen, A. Asai, V. Zhong, R. Das, D. Khashabi, and H. Hajishirzi, “When not to trust
447 language models: Investigating effectiveness of parametric and non-parametric memories,” in
448 *Annual Meeting of the Association for Computational Linguistics*, pp. 9802–9822, 2023.
- 449 [36] X. Ho, A.-K. D. Nguyen, S. Sugawara, and A. Aizawa, “Constructing a multi-hop qa dataset for
450 comprehensive evaluation of reasoning steps,” in *International Conference on Computational*
451 *Linguistics*, pp. 6609–6625, 2020.
- 452 [37] OpenAI, “gpt-oss-120b.” <https://huggingface.co/openai/gpt-oss-120b>, 2025.
- 453 [38] R. S. Sutton and A. G. Barto, *Reinforcement learning: An introduction*, vol. 1. MIT press
454 Cambridge, 1998.
- 455 [39] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, “Soft actor-critic: Off-policy maximum entropy
456 deep reinforcement learning with a stochastic actor,” in *International Conference on Machine*
457 *Learning*, pp. 1861–1870, 2018.
- 458 [40] Y. Nesterov, *Introductory lectures on convex optimization: A basic course*, vol. 87. Springer
459 Science & Business Media, 2013.
- 460 [41] S. Bubeck, “Convex optimization: Algorithms and complexity,” *Foundations and Trends® in*
461 *Machine Learning*, vol. 8, no. 3-4, pp. 231–357, 2015.
- 462 [42] S. Zeng, C. Li, A. Garcia, and M. Hong, “Maximum-likelihood inverse reinforcement learning
463 with finite-time guarantees,” *Advances in Neural Information Processing Systems*, vol. 35,
464 pp. 10122–10135, 2022.

A Adaptation to Non-binary Outcome Rewards

While this paper considers the case where the outcome reward is a binary success reward, the proposed framework can be directly applied to cases where the outcome reward is non-binary. In our framework, we need the outcome reward in the upper level (3) to learn a shaping reward. The upper-level loss consists of a cross-entropy (CE) loss and a DPO loss.

The CE loss also supports non-binary outcome rewards as long as the outcome reward $r_{\text{out}}(x, y) \in [0, 1]$. In this case, the CE loss becomes:

$$L_{\text{ce}}(\theta, \pi_{r_\theta}) \triangleq -E_{x \sim \mathcal{D}, y \sim \pi_{r_\theta}(\cdot|x)} \left[r_{\text{out}}(x, y) \cdot \log \sigma(r_\theta(x, y)) + (1 - r_{\text{out}}(x, y)) \cdot \log[1 - \sigma(r_\theta(x, y))] \right].$$

In the case where the original outcome reward is beyond $[0, 1]$, e.g., $r_{\text{out}}(x, y) \in [C_l, C_r]$ where C_l and C_r are some bounded constants, we can normalize the outcome reward to $[0, 1]$ and get the normalized outcome reward $\bar{r}_{\text{out}} = \frac{r_{\text{out}} - C_l}{C_r - C_l}$. We can use the normalized outcome reward in the CE loss. Note that $\bar{r}_{\text{out}}$ and r_{out} have the same optimal policy because any linear transformation of the reward function does not change the corresponding optimal policy [13].

The DPO loss also works for the non-binary outcome reward because we can still formulate preference pairs where a trajectory with a higher outcome reward is preferred to a trajectory with a lower outcome reward.

B Proofs

In this section, we provide the theoretical foundations and proof for the theoretical statements in the paper.

In RL [38], the policy is stationary for infinite-horizon MDPs, while the policy should depend on t for finite-horizon MDPs. This leads to a mismatch for agentic RL and RL for LLMs. Specifically, the generation of LLMs is always finite-horizon (finite number of tokens or finite number of turns), which fits finite-horizon MDPs and thus the policy model should be time-dependent. However, in practice, only a stationary LLM policy is learned. To address this issue, in the proof, we consider a unified infinite-horizon MDP $(\mathcal{S}, \mathcal{A}, \mathcal{D}, P, \gamma, r)$ that does not include the time horizon T , and its time horizon is infinite. However, we treat s_{T+1} as an absorbing state, i.e., it can only transit to itself regardless of actions. Moreover, the reward $r(s_t, a_t) \equiv 0$ for $t > T$. By this way, we use an infinite-horizon MDP to model the finite-horizon MDP in the paper. The discount factor is $\gamma \in (0, 1)$ for infinite-horizon MDPs. Moreover, the next state is $s_{t+1} = (s_t, a_t, f_t)$ where f_t is the environment feedback that depends on the action a_t . In the proof, for simple notations, we omit f_t and write $s_{t+1} = (s_t, a_t)$.

Lemma 3 ([29, 39]). *The optimal policy π_{r_θ} of the lower-level optimization problem (4) is the soft policy:*

$$\begin{aligned} \pi_{r_\theta}(a_t|s_t) &= \frac{\pi_{\text{ref}}(a_t|s_t) \exp(\frac{1}{\beta} Q_{r_\theta}^*(s_t, a_t))}{\sum_{a' \in \mathcal{A}} \pi_{\text{ref}}(a'|s_t) \exp(\frac{1}{\beta} Q_{r_\theta}^*(s_t, a'))} \\ Q_{r_\theta}^*(s_t, a_t) &= r_\theta(s_t, a_t) + \gamma V_{r_\theta}^*(s_{t+1}) \\ V_{r_\theta}^*(s_t) &= \beta \log \left[\sum_{a' \in \mathcal{A}} \pi_{\text{ref}}(a'|s_t) \exp(\frac{1}{\beta} Q_{r_\theta}^*(s_t, a')) \right], \end{aligned}$$

where $Q_{r_\theta}^*$ and $V_{r_\theta}^*$ are respectively the optimal action-value function and state-value function under the reward r_θ .

B.1 Proof of Lemma 1

The full hypergradient needs to consider the implicit dependency of the policy π_{r_θ} on θ . Therefore, the following claim computes the gradient $\nabla_\theta \log \pi_{r_\theta}$.

Claim 1. *We have the following gradient:*

$$\begin{aligned} &\nabla_\theta \log \pi_{r_\theta}(a_t|s_t) \\ &= \frac{1}{\beta} E^{\pi_{r_\theta}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_\theta r_\theta(S_i, A_i) | S_t = s_t, A_t = a_t \right] - \frac{1}{\beta} E^{\pi_{r_\theta}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_\theta r_\theta(S_i, A_i) | S_t = s_t \right]. \end{aligned}$$

502 *Proof.* Define $Z_{r_\theta}(s_t, a_t) \triangleq \pi_{\text{ref}}(a_t|s_t) \exp(\frac{1}{\beta} Q_{r_\theta}^*(s_t, a_t))$ and $Z_{r_\theta}(s_t) \triangleq \exp(\frac{1}{\beta} V_{r_\theta}^*(s_t))$.

$$\begin{aligned}
\nabla_\theta \log Z_{r_\theta}(s_t) &= \frac{\sum_{a_t \in \mathcal{A}} \nabla_\theta Z_{r_\theta}(s_t, a_t)}{Z_{r_\theta}(s_t)}, \\
&= \sum_{a_t \in \mathcal{A}} \frac{Z_{r_\theta}(s_t, a_t)}{Z_{r_\theta}(s_t)} \nabla_\theta \log Z_{r_\theta}(s_t, a_t), \\
&= \sum_{a_t \in \mathcal{A}} \pi_{r_\theta}(a_t|s_t) \left[\frac{\nabla_\theta r_\theta(s_t, a_t)}{\beta} + \gamma \nabla_\theta \log Z_{r_\theta}(s_{t+1}) \right], \\
&= \sum_{a_t \in \mathcal{A}} \pi_{r_\theta}(a_t|s_t) \left[\frac{\nabla_\theta r_\theta(s_t, a_t)}{\beta} \right. \\
&\quad \left. + \gamma \sum_{a_{t+1} \in \mathcal{A}} \pi_{r_\theta}(a_{t+1}|s_{t+1}) \left(\frac{\nabla_\theta r_\theta(s_{t+1}, a_{t+1})}{\beta} + \gamma \nabla_\theta \log Z_{r_\theta}(s_{t+2}) \right) \right].
\end{aligned}$$

By continuing to expand $\nabla_\theta \log Z_{r_\theta}(s_{t+2})$, we can get

$$\nabla_\theta \log Z_{r_\theta}(s_t) = \frac{1}{\beta} E^{\pi_{r_\theta}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_\theta r_\theta(S_i, A_i) | S_t = s_t \right],$$

and similarly we can get

$$\nabla_\theta \log Z_{r_\theta}(s_t, a_t) = \frac{1}{\beta} E^{\pi_{r_\theta}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_\theta r_\theta(S_i, A_i) | S_t = s_t, A_t = a_t \right].$$

503 Thus, we have the gradient $\nabla_\theta \log \pi_{r_\theta}(a_t|s_t) = \nabla_\theta \log Z_{r_\theta}(s_t, a_t) - \nabla_\theta \log Z_{r_\theta}(s_t)$. □

504 Now, we can proceed to obtain the hypergradient. Define the pointwise cross-entropy loss

$$l_{\text{ce}}(\theta; (x, y)) \triangleq - \left[r_{\text{succ}}(x, y) \log \sigma(r_\theta(x, y)) + (1 - r_{\text{succ}}(x, y)) \log (1 - \sigma(r_\theta(x, y))) \right].$$

505 Then, the overall loss is

$$L_{\text{ce}}(\theta, \pi_{r_\theta}) \triangleq E_{x \sim \mathcal{D}, y \sim \pi_{r_\theta}(\cdot|x)} [l_{\text{ce}}(\theta; (x, y))].$$

506 To compute the full gradient with respect to θ , we differentiate both the explicit dependence of l_{ce} on
507 θ and the implicit dependence through $y \sim \pi_{r_\theta}(\cdot|x)$:

$$\begin{aligned}
\nabla_\theta L_{\text{ce}}(\theta, \pi_{r_\theta}) &= \nabla_\theta E_{x \sim \mathcal{D}, y \sim \pi_{r_\theta}(\cdot|x)} [l_{\text{ce}}(\theta; (x, y))] \\
&= E_{x \sim \mathcal{D}, y \sim \pi_{r_\theta}(\cdot|x)} \left[l_{\text{ce}}(\theta; (x, y)) \nabla_\theta \log \pi_{r_\theta}(y|x) + \nabla_\theta l_{\text{ce}}(\theta; (x, y)) \right]. \quad (5)
\end{aligned}$$

508 Recall from Claim 1 that

$$\begin{aligned}
\nabla_\theta \log Z_{r_\theta}(s_t, a_t) &= \frac{1}{\beta} E^{\pi_{r_\theta}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_\theta r_\theta(S_i, A_i) | S_t = s_t, A_t = a_t \right] \\
\nabla_\theta \log Z_{r_\theta}(s_t) &= \frac{1}{\beta} E^{\pi_{r_\theta}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_\theta r_\theta(S_i, A_i) | S_t = s_t \right].
\end{aligned}$$

509 To precisely compute the hypergradient (5), at each state s_t , we need to sample multiple trajectories,

510 e.g., sample m trajectories $\{s_t, a_t^j, s_{t+1}^j, \dots, s_T^j, a_T^j, \dots\}_{j=1}^m$. Then we can estimate

$$\begin{aligned}
\nabla_\theta \log Z_{r_\theta}(s_t, a_t^j) &\approx \frac{1}{\beta} \left[\nabla_\theta r_\theta(s_t, a_t^j) + \sum_{i=t+1}^{\infty} \gamma^{i-t} \nabla_\theta r_\theta(s_i^j, a_i^j) \right] \\
\nabla_\theta \log Z_{r_\theta}(s_t) &\approx \frac{1}{\beta} \left[\frac{1}{m} \sum_{j=1}^m \nabla_\theta r_\theta(s_t, a_t^j) + \sum_{i=t+1}^{\infty} \gamma^{i-t} \nabla_\theta r_\theta(s_i^j, a_i^j) \right].
\end{aligned}$$

However, it is intractable to estimate $\nabla_{\theta} \log Z_{r_{\theta}}(s_t, a_t^j)$ and $\nabla_{\theta} \log Z_{r_{\theta}}(s_t)$ at each time t in this way because it requires to sample m trajectories at each time t . To address this issue, we use an approximation method proposed in subsection 4.1.3 of GRPO [15]. Specifically, we only sample m trajectories at the initial state $s_1^{(n)} = x^{(n)}: \{s_1^{(n,j)}, a_1^{(n,j)}, \dots, s_T^{(n,j)}, a_T^{(n,j)}, \dots\}_{j=1}^m$. We approximate $\nabla_{\theta} \log Z_{r_{\theta}}(S_t)$ via

$$\nabla_{\theta} \log Z_{r_{\theta}}(S_t) \approx \frac{1}{\beta m} \sum_{j'=1}^m \sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(s_i^{(n,j')}, a_i^{(n,j')}),$$

511 therefore, we can approximate the gradient $\nabla_{\theta} \log \pi_{r_{\theta}}(y^{(n,j)} | x^{(n)})$ via

$$\begin{aligned} \nabla_{\theta} \log \pi_{r_{\theta}}(y^{(n,j)} | x^{(n)}) &= \sum_{t=1}^{\infty} \nabla_{\theta} \log \pi_{r_{\theta}}(a_t^{(n,j)} | s_t^{(n,j)}) \\ &\approx \sum_{t=1}^{\infty} \left(\frac{1}{\beta} \sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(s_i^{(n,j)}, a_i^{(n,j)}) - \frac{1}{\beta m} \sum_{j'=1}^m \sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(s_i^{(n,j')}, a_i^{(n,j')}) \right). \end{aligned} \quad (6)$$

512 By plugging (6) into (5), we can approximate the CE hypergradient $\nabla L_{\text{ce}}(\theta, \pi_{r_{\theta}})$ via $g_{\text{ce}}(\theta)$:

$$\begin{aligned} g_{\text{ce}}(\theta) &= \frac{1}{Bm} \sum_{n=1}^B \sum_{j=1}^m \left[\nabla_{\theta} l_{\text{ce}}(\theta; (x^{(n)}, y^{(n,j)})) + l_{\text{ce}}(\theta; (x^{(n)}, y^{(n,j)})) \right. \\ &\quad \left. \sum_{t=1}^{\infty} \left(\frac{1}{\beta} \sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(s_i^{(n,j)}, a_i^{(n,j)}) - \frac{1}{\beta m} \sum_{j'=1}^m \sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(s_i^{(n,j')}, a_i^{(n,j')}) \right) \right]. \end{aligned} \quad (7)$$

513 For simple notations, we denote

$$\begin{aligned} \Delta_{\text{ce}}^{(n,j)} &= \sum_{t=1}^{\infty} \left(\frac{1}{\beta} \sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(s_i^{(n,j)}, a_i^{(n,j)}) - \frac{1}{\beta m} \sum_{j'=1}^m \sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(s_i^{(n,j')}, a_i^{(n,j')}) \right) \\ &\stackrel{(a)}{=} \sum_{t=1}^T \left(\frac{1}{\beta} \sum_{i=t}^T \gamma^{i-t} \nabla_{\theta} r_{\theta}(s_i^{(n,j)}, a_i^{(n,j)}) - \frac{1}{\beta m} \sum_{j'=1}^m \sum_{i=t}^T \gamma^{i-t} \nabla_{\theta} r_{\theta}(s_i^{(n,j')}, a_i^{(n,j')}) \right), \end{aligned}$$

514 where (a) follows from the fact that $r_{\theta}(s_t, a_t) \equiv 0$ for $t > T$ in our infinite-time MDP formulation.

515 Therefore,

$$g_{\text{ce}}(\theta) = \frac{1}{Bm} \sum_{n=1}^B \sum_{j=1}^m \left[\nabla_{\theta} l_{\text{ce}}(\theta; (x^{(n)}, y^{(n,j)})) + l_{\text{ce}}(\theta; (x^{(n)}, y^{(n,j)})) \cdot \Delta_{\text{ce}}^{(n,j)} \right].$$

516 The first term in (7) is the standard gradient of a CE loss $\nabla_{\theta} l_{\text{ce}}(\theta; (x^{(n)}, y^{(n,j)}))$. For the sake of
517 completeness, we also include its expression here. Let

$$s \triangleq r_{\text{succ}}(x, y), \quad z_{\theta} \triangleq r_{\theta}(x, y).$$

518 Then

$$l_{\text{ce}}(\theta; (x, y)) = - \left[s \log \sigma(z_{\theta}) + (1 - s) \log(1 - \sigma(z_{\theta})) \right].$$

519 Differentiating with respect to θ ,

$$\nabla_{\theta} l_{\text{ce}}(\theta; (x, y)) = - [s \nabla_{\theta} \log \sigma(z_{\theta}) + (1 - s) \nabla_{\theta} \log(1 - \sigma(z_{\theta}))].$$

520 Using

$$\nabla_{\theta} \log \sigma(z_{\theta}) = (1 - \sigma(z_{\theta})) \nabla_{\theta} z_{\theta}, \quad \nabla_{\theta} \log(1 - \sigma(z_{\theta})) = -\sigma(z_{\theta}) \nabla_{\theta} z_{\theta},$$

521 we obtain

$$\nabla_{\theta} l_{\text{ce}}(\theta; (x, y)) = (\sigma(z_{\theta}) - s) \nabla_{\theta} z_{\theta} = (\sigma(r_{\theta}(x, y)) - r_{\text{succ}}(x, y)) \nabla_{\theta} r_{\theta}(x, y). \quad (8)$$

522 B.2 Proof of Lemma 2

523 Consider a prompt x together with a preferred response y^+ and a dispreferred response y^- . Define
 524 the reward-induced preference margin

$$\Delta_\theta(x, y^+, y^-) \triangleq r_\theta(x, y^+) - r_\theta(x, y^-),$$

525 and the pointwise DPO loss

$$l_{\text{dpo}}(\theta; (x, y^+, y^-)) \triangleq -\log \sigma(\Delta_\theta(x, y^+, y^-)).$$

526 The overall DPO loss is

$$L_{\text{dpo}}(\theta, \pi_{r_\theta}) \triangleq E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_{r_\theta}(\cdot, \cdot | x)} [l_{\text{dpo}}(\theta; (x, y^+, y^-))].$$

527 Again, the full gradient must account for both the explicit dependence of l_{dpo} on θ and the implicit
 528 dependence through sampling from π_{r_θ} :

$$\begin{aligned} \nabla_\theta L_{\text{dpo}}(\theta, \pi_{r_\theta}) &= E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_{r_\theta}(\cdot, \cdot | x)} \left[l_{\text{dpo}}(\theta; (x, y^+, y^-)) \nabla_\theta \log \pi_{r_\theta}(y^+, y^- | x) + \nabla_\theta l_{\text{dpo}}(\theta; (x, y^+, y^-)) \right]. \end{aligned} \quad (9)$$

529 Since the pair is formed by independent samples from the same policy, then

$$\log \pi_{r_\theta}(y^+, y^- | x) = \log \pi_{r_\theta}(y^+ | x) + \log \pi_{r_\theta}(y^- | x),$$

530 so

$$\nabla_\theta \log \pi_{r_\theta}(y^+, y^- | x) = \nabla_\theta \log \pi_{r_\theta}(y^+ | x) + \nabla_\theta \log \pi_{r_\theta}(y^- | x). \quad (10)$$

531 By plugging (6) and (10) into (9), we can approximate the hypergradient $\nabla_\theta L_{\text{dpo}}(\theta, \pi_{r_\theta})$ via $g_{\text{dpo}}(\theta)$:

$$\begin{aligned} g_{\text{dpo}}(\theta) &= \frac{1}{B} \sum_{n=1}^B \frac{1}{\bar{m}^{(n)}(m - \bar{m}^{(n)})} \sum_{\bar{j}=1}^{\bar{m}^{(n)}(m - \bar{m}^{(n)})} \left[\nabla_\theta l_{\text{dpo}}(\theta; (x^{(n)}, y^{(n, \bar{j}, +)}, y^{(n, \bar{j}, -)})) \right. \\ &\quad + l_{\text{dpo}}(\theta; (x^{(n)}, y^{(n, \bar{j}, +)}, y^{(n, \bar{j}, -)})) \cdot \sum_{t=1}^{\infty} \left(\frac{1}{\beta} \sum_{i=t}^{\infty} \gamma^{i-t} \nabla_\theta r_\theta(s_i^{(n, \bar{j}, +)}, a_i^{(n, \bar{j}, +)}) \right. \\ &\quad \left. \left. + \frac{1}{\beta} \sum_{i=t}^{\infty} \gamma^{i-t} \nabla_\theta r_\theta(s_i^{(n, \bar{j}, -)}, a_i^{(n, \bar{j}, -)}) - \frac{2}{\beta m} \sum_{j=1}^m \sum_{i=t}^{\infty} \gamma^{i-t} \nabla_\theta r_\theta(s_i^{(n, j)}, a_i^{(n, j)}) \right) \right]. \end{aligned} \quad (11)$$

532 For simple notations, we denote

$$\begin{aligned} \Delta_{\text{dpo}}^{(n, \bar{j})} &= \sum_{t=1}^{\infty} \left(\frac{1}{\beta} \sum_{i=t}^{\infty} \gamma^{i-t} \nabla_\theta r_\theta(s_i^{(n, \bar{j}, +)}, a_i^{(n, \bar{j}, +)}) \right. \\ &\quad \left. + \frac{1}{\beta} \sum_{i=t}^{\infty} \gamma^{i-t} \nabla_\theta r_\theta(s_i^{(n, \bar{j}, -)}, a_i^{(n, \bar{j}, -)}) - \frac{2}{\beta m} \sum_{j=1}^m \sum_{i=t}^{\infty} \gamma^{i-t} \nabla_\theta r_\theta(s_i^{(n, j)}, a_i^{(n, j)}) \right) \\ &\stackrel{(a)}{=} \sum_{t=1}^T \left(\frac{1}{\beta} \sum_{i=t}^T \gamma^{i-t} \nabla_\theta r_\theta(s_i^{(n, \bar{j}, +)}, a_i^{(n, \bar{j}, +)}) \right. \\ &\quad \left. + \frac{1}{\beta} \sum_{i=t}^T \gamma^{i-t} \nabla_\theta r_\theta(s_i^{(n, \bar{j}, -)}, a_i^{(n, \bar{j}, -)}) - \frac{2}{\beta m} \sum_{j=1}^m \sum_{i=t}^{\infty} \gamma^{i-t} \nabla_\theta r_\theta(s_i^{(n, j)}, a_i^{(n, j)}) \right), \end{aligned}$$

533 where (a) follows from the fact that $r_\theta(s_t, a_t) \equiv 0$ for $t > T$ in our infinite-time MDP formulation.

534 Therefore,

$$g_{\text{dpo}}(\theta) = \frac{1}{B} \sum_{n=1}^B \frac{1}{\bar{m}^{(n)}(m - \bar{m}^{(n)})} \sum_{\bar{j}=1}^{\bar{m}^{(n)}(m - \bar{m}^{(n)})} \left[\nabla_\theta l_{\text{dpo}}(\theta; (x^{(n)}, y^{(n, \bar{j}, +)}, y^{(n, \bar{j}, -)})) \right]$$

$$+ l_{\text{dpo}}(\theta; (x^{(n)}, y^{(n, \bar{j}, +)}, y^{(n, \bar{j}, -)})) \cdot \Delta_{\text{dpo}}^{(n, \bar{j})}.$$

535 The first term in (11) is the standard gradient of a DPO loss $\nabla_{\theta} l_{\text{dpo}}(\theta; (x^{(n)}, y^{(n, \bar{j}, +)}, y^{(n, \bar{j}, -)}))$. For
 536 the sake of completeness, we compute the gradient of l_{dpo} . Since

$$l_{\text{dpo}}(\theta; (x, y^+, y^-)) = -\log \sigma(\Delta_{\theta}),$$

537 we have

$$\nabla_{\theta} l_{\text{dpo}}(\theta; (x, y^+, y^-)) = -(1 - \sigma(\Delta_{\theta})) \nabla_{\theta} \Delta_{\theta}.$$

538 Using

$$\Delta_{\theta}(x, y^+, y^-) = r_{\theta}(x, y^+) - r_{\theta}(x, y^-),$$

539 its gradient is

$$\nabla_{\theta} \Delta_{\theta}(x, y^+, y^-) = \nabla_{\theta} r_{\theta}(x, y^+) - \nabla_{\theta} r_{\theta}(x, y^-).$$

540 Therefore,

$$\nabla_{\theta} l_{\text{dpo}}(\theta; (x, y^+, y^-)) = -(1 - \sigma(\Delta_{\theta}(x, y^+, y^-))) (\nabla_{\theta} r_{\theta}(x, y^+) - \nabla_{\theta} r_{\theta}(x, y^-)). \quad (12)$$

541 B.3 Preliminary Theoretical Results for Theorem 1

542 **Lemma 4.** Suppose Assumption 1 holds, then the shaping reward function r_{θ} satisfies
 543 $\|\nabla_{\theta} r_{\theta}(s, a)\| \leq C$ and $\|\nabla_{\theta\theta}^2 r_{\theta}(s, a)\| \leq \bar{C}$ for any $(s, a) \in \mathcal{S} \times \mathcal{A}$ where C and $\bar{C}$ are posi-
 544 tive constants.

545 *Proof.* This is a direct result based on Assumption 1. From the potential-based parameteriza-
 546 tion, we can see that $\nabla_{\theta} r_{\theta}(s_t, a_t) = \gamma^{1-t} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$ and $\nabla_{\theta\theta}^2 r_{\theta}(s_t, a_t) = \nabla_{\theta\theta}^2 \log \pi_{\theta}(a_t | s_t)$.
 547 Therefore, $\|\nabla_{\theta} r_{\theta}(s, a)\| \leq \gamma^{1-T} \|\nabla_{\theta} \log \pi_{\theta}(a | s)\|$ and $\|\nabla_{\theta\theta}^2 r_{\theta}(s, a)\| \leq \gamma^{1-T} \|\nabla_{\theta\theta}^2 \log \pi_{\theta}(a | s)\|$.
 548 Since γ^{1-T} is a bounded constant, we can always find C to upper bound both $\|\nabla_{\theta} r_{\theta}(s, a)\|$ and
 549 $\|\nabla_{\theta} \log \pi_{\theta}(a_t | s_t)\|$, and find $\bar{C}$ to upper bound both $\|\nabla_{\theta\theta}^2 r_{\theta}(s, a)\|$ and $\|\nabla_{\theta\theta}^2 \log \pi_{\theta}(a | s)\|$. For
 550 simplicity, we use C and $\bar{C}$ to bound both. $\square$

551 **Lemma 5.** Suppose Assumption 1 holds, then it holds that

$$\begin{aligned} 0 &\leq l_{\text{ce}}(\theta; (x, y)) \leq \log(1 + e^{T\bar{C}}) \\ \|\nabla_{\theta} l_{\text{ce}}(\theta; (x, y))\| &\leq TC \\ \|\nabla_{\theta\theta}^2 l_{\text{ce}}(\theta; (x, y))\| &\leq \frac{1}{4} T^2 C^2 + T\bar{C}. \end{aligned}$$

552 *Proof.* Recall that

$$l_{\text{ce}}(\theta; (x, y)) = -\left[r_{\text{succ}}(x, y) \log \sigma(r_{\theta}(x, y)) + (1 - r_{\text{succ}}(x, y)) \log (1 - \sigma(r_{\theta}(x, y))) \right].$$

553 Let

$$s \triangleq r_{\text{succ}}(x, y) \in \{0, 1\}, \quad z_{\theta} \triangleq r_{\theta}(x, y).$$

554 Then

$$l_{\text{ce}}(\theta; (x, y)) = -[s \log \sigma(z_{\theta}) + (1 - s) \log (1 - \sigma(z_{\theta}))].$$

555 We first rewrite l_{ce} as a scalar function of z_{θ} . Define

$$\phi_s(z) \triangleq -[s \log \sigma(z) + (1 - s) \log (1 - \sigma(z))].$$

556 Then

$$l_{\text{ce}}(\theta; (x, y)) = \phi_s(z_{\theta}).$$

557 **Step 1: boundedness of the loss.** Since $s \in \{0, 1\}$, there are two cases.

If $s = 1$, then

$$\phi_1(z) = -\log \sigma(z) = \log(1 + e^{-z}).$$

Since $r_\theta(s, a) \leq \tilde{C}$, then $|z_\theta| \leq T\tilde{C}$, we have $z_\theta \in [-T\tilde{C}, T\tilde{C}]$, and thus

$$0 \leq \phi_1(z_\theta) \leq \phi_1(-T\tilde{C}) = \log(1 + e^{T\tilde{C}}).$$

If $s = 0$, then

$$\phi_0(z) = -\log(1 - \sigma(z)) = \log(1 + e^z),$$

so again using $z_\theta \in [-T\tilde{C}, T\tilde{C}]$,

$$0 \leq \phi_0(z_\theta) \leq \phi_0(T\tilde{C}) = \log(1 + e^{T\tilde{C}}).$$

Therefore, in both cases,

$$0 \leq l_{\text{ce}}(\theta; (x, y)) \leq \log(1 + e^{T\tilde{C}}).$$

Step 2: boundedness of the gradient. The derivative of ϕ_s is

$$\phi'_s(z) = \sigma(z) - s.$$

Hence, by the chain rule,

$$\nabla_\theta l_{\text{ce}}(\theta; (x, y)) = \phi'_s(z_\theta) \nabla_\theta z_\theta = (\sigma(z_\theta) - s) \nabla_\theta r_\theta(x, y).$$

Since $s \in \{0, 1\}$ and $\sigma(z_\theta) \in (0, 1)$,

$$|\sigma(z_\theta) - s| \leq 1.$$

558 Therefore,

$$\|\nabla_\theta l_{\text{ce}}(\theta; (x, y))\| \leq |\sigma(z_\theta) - s| \|\nabla_\theta r_\theta(x, y)\| \leq TC.$$

559 **Step 3: boundedness of the Hessian.** The second derivative of ϕ_s is

$$\phi''_s(z) = \sigma(z)(1 - \sigma(z)).$$

560 Thus,

$$0 \leq \phi''_s(z) \leq \frac{1}{4}.$$

561 Differentiating once more and applying the chain rule gives

$$\nabla_{\theta\theta}^2 l_{\text{ce}}(\theta; (x, y)) = \phi''_s(z_\theta) \nabla_\theta z_\theta \nabla_\theta z_\theta^\top + \phi'_s(z_\theta) \nabla_{\theta\theta}^2 r_\theta.$$

562 Substituting back $z_\theta = r_\theta(x, y)$,

$$\nabla_{\theta\theta}^2 l_{\text{ce}}(\theta; (x, y)) = \sigma(z_\theta)(1 - \sigma(z_\theta)) \nabla_\theta r_\theta(x, y) \nabla_\theta r_\theta(x, y)^\top + (\sigma(z_\theta) - s) \nabla_{\theta\theta}^2 r_\theta(x, y).$$

563 Taking norms and using the triangle inequality,

$$\begin{aligned} \|\nabla_{\theta\theta}^2 l_{\text{ce}}(\theta; (x, y))\| &\leq \sigma(z_\theta)(1 - \sigma(z_\theta)) \|\nabla_\theta r_\theta(x, y) \nabla_\theta r_\theta(x, y)^\top\| + |\sigma(z_\theta) - s| \|\nabla_{\theta\theta}^2 r_\theta(x, y)\| \\ &\leq \frac{1}{4} \|\nabla_\theta r_\theta(x, y)\|^2 + \bar{C} \\ &\leq \frac{1}{4} T^2 C^2 + T\bar{C}. \end{aligned}$$

564 Here we use

$$\|\nabla_\theta r_\theta(x, y) \nabla_\theta r_\theta(x, y)^\top\| = \|\nabla_\theta r_\theta(x, y)\|^2$$

565 for the operator norm. □

566 **Lemma 6.** Suppose Assumption 1 holds, then it holds that

$$\begin{aligned} 0 \leq l_{\text{dpo}}(\theta; (x, y^+, y^-)) &\leq \log(1 + e^{2T\tilde{C}}) \\ \|\nabla_\theta l_{\text{dpo}}(\theta; (x, y^+, y^-))\| &\leq 2TC \\ \|\nabla_{\theta\theta}^2 l_{\text{dpo}}(\theta; (x, y^+, y^-))\| &\leq T^2 C^2 + 2T\bar{C}. \end{aligned}$$

567 *Proof.* Recall that

$$\Delta_\theta(x, y^+, y^-) = r_\theta(x, y^+) - r_\theta(x, y^-), \quad l_{\text{dpo}}(\theta; (x, y^+, y^-)) = -\log \sigma(\Delta_\theta(x, y^+, y^-)).$$

568 Define

$$\phi(z) \triangleq -\log \sigma(z) = \log(1 + e^{-z}),$$

569 so that

$$l_{\text{dpo}}(\theta; (x, y^+, y^-)) = \phi(\Delta_\theta).$$

570 **Step 1: bounds on Δ_θ .** By the assumptions on r_θ ,

$$|\Delta_\theta| \leq |r_\theta(x, y^+)| + |r_\theta(x, y^-)| \leq 2T\tilde{C},$$

$$\|\nabla_\theta \Delta_\theta\| = \|\nabla_\theta r_\theta(x, y^+) - \nabla_\theta r_\theta(x, y^-)\| \leq 2TC,$$

572 and

$$\|\nabla_{\theta\theta}^2 \Delta_\theta\| = \|\nabla_{\theta\theta}^2 r_\theta(x, y^+) - \nabla_{\theta\theta}^2 r_\theta(x, y^-)\| \leq 2T\bar{C}.$$

573 **Step 2: boundedness of the loss.** Since $\phi(z) = \log(1 + e^{-z})$ and $\Delta_\theta \in [-2T\tilde{C}, 2T\tilde{C}]$, we have

$$0 \leq \phi(\Delta_\theta) \leq \phi(-2T\tilde{C}) = \log(1 + e^{2T\tilde{C}}).$$

574 Therefore,

$$0 \leq l_{\text{dpo}}(\theta; (x, y^+, y^-)) \leq \log(1 + e^{2T\tilde{C}}).$$

575 **Step 3: boundedness of the gradient.** The derivative of ϕ is

$$\phi'(z) = \sigma(z) - 1.$$

576 Hence, by the chain rule,

$$\nabla_\theta l_{\text{dpo}} = \phi'(\Delta_\theta) \nabla_\theta \Delta_\theta = (\sigma(\Delta_\theta) - 1) \nabla_\theta \Delta_\theta.$$

577 Since $|\sigma(\Delta_\theta) - 1| \leq 1$,

$$\|\nabla_\theta l_{\text{dpo}}\| \leq \|\nabla_\theta \Delta_\theta\| \leq 2TC.$$

578 **Step 4: boundedness of the Hessian.** The second derivative of ϕ is

$$\phi''(z) = \sigma(z)(1 - \sigma(z)), \quad 0 \leq \phi''(z) \leq \frac{1}{4}.$$

579 Applying the chain rule again,

$$\nabla_{\theta\theta}^2 l_{\text{dpo}} = \phi''(\Delta_\theta) \nabla_\theta \Delta_\theta \nabla_\theta \Delta_\theta^\top + \phi'(\Delta_\theta) \nabla_{\theta\theta}^2 \Delta_\theta.$$

580 Therefore,

$$\begin{aligned} \|\nabla_{\theta\theta}^2 l_{\text{dpo}}\| &\leq \phi''(\Delta_\theta) \|\nabla_\theta \Delta_\theta \nabla_\theta \Delta_\theta^\top\| + |\phi'(\Delta_\theta)| \|\nabla_{\theta\theta}^2 \Delta_\theta\| \\ &\leq \frac{1}{4} \|\nabla_\theta \Delta_\theta\|^2 + \|\nabla_{\theta\theta}^2 \Delta_\theta\| \\ &\leq \frac{1}{4} (2TC)^2 + 2T\bar{C} \\ &= T^2 C^2 + 2T\bar{C}. \end{aligned}$$

581

□

582 **Lemma 7.** Suppose Assumption 1 holds, then it holds that $\|\nabla L(\theta, \pi_{r_\theta})\| \leq (1 +$
583 $2\lambda)TC + \frac{[2\log(1+e^{T\tilde{C}})+4\lambda\log(1+e^{2T\tilde{C}})]T^2C}{\beta}$ and $\|g_{\text{ce}}(\theta) + \lambda g_{\text{dpo}}(\theta)\| \leq (1 + 2\lambda)TC +$
584 $\frac{[2\log(1+e^{T\tilde{C}})+4\lambda\log(1+e^{2T\tilde{C}})]T^2C}{\beta}$ for any θ .

585 *Proof.* Recall from Lemma 1 that

$$\begin{aligned} g_{\text{ce}}(\theta) &= \frac{1}{Bm} \sum_{n=1}^B \sum_{j=1}^m \left[\nabla_\theta l_{\text{ce}}(\theta; (x^{(n)}, y^{(n,j)})) + l_{\text{ce}}(\theta; (x^{(n)}, y^{(n,j)})) \right. \\ &\quad \left. \sum_{t=1}^T \left(\frac{1}{\beta} \sum_{i=t}^T \gamma^{i-t} \nabla_\theta r_\theta(s_i^{(n,j)}, a_i^{(n,j)}) - \frac{1}{\beta m} \sum_{j'=1}^m \sum_{i=t}^T \gamma^{i-t} \nabla_\theta r_\theta(s_i^{(n,j')}, a_i^{(n,j')}) \right) \right]. \end{aligned}$$

586 Since $|l_{\text{ce}}(\theta; (x, y))| \leq \log(1 + e^{T\tilde{C}})$ and $\|\nabla_\theta l_{\text{ce}}(\theta; (x, y))\| \leq TC$ (Lemma 5) and $\|\nabla_\theta r(s, a)\| \leq$
587 C (Lemma 4), we can easily see that $\|g_{\text{ce}}(\theta)\| \leq TC + \frac{2\log(1+e^{T\tilde{C}})T^2C}{\beta}$.

588 Recall from Lemma 2 that

$$\begin{aligned}
g_{\text{dpo}}(\theta) &= \frac{1}{B} \sum_{n=1}^B \frac{1}{\bar{m}^{(n)}(m - \bar{m}^{(n)})} \sum_{j=1}^{\bar{m}^{(n)}(m - \bar{m}^{(n)})} \left[\nabla_{\theta} l_{\text{dpo}}(\theta; (x^{(n)}, y^{(n, \bar{j}, +)}, y^{(n, \bar{j}, -)})) \right. \\
&\quad + l_{\text{dpo}}(\theta; (x^{(n)}, y^{(n, \bar{j}, +)}, y^{(n, \bar{j}, -)})) \cdot \sum_{t=1}^T \left(\frac{1}{\beta} \sum_{i=t}^T \gamma^{i-t} \nabla_{\theta} r_{\theta}(s_i^{(n, \bar{j}, +)}, a_i^{(n, \bar{j}, +)}) \right. \\
&\quad \left. \left. + \frac{1}{\beta} \sum_{i=t}^T \gamma^{i-t} \nabla_{\theta} r_{\theta}(s_i^{(n, \bar{j}, -)}, a_i^{(n, \bar{j}, -)}) - \frac{2}{\beta m} \sum_{j=1}^m \sum_{i=t}^T \gamma^{i-t} \nabla_{\theta} r_{\theta}(s_i^{(n, j)}, a_i^{(n, j)}) \right) \right].
\end{aligned}$$

589 Since $|l_{\text{dpo}}(\theta, (x, y^+, y^-))| \leq \log(1 + e^{2T\bar{C}})$ and $\|\nabla_{\theta} l_{\text{dpo}}(\theta, (x, y^+, y^-))\| \leq 2TC$ (Lemma 6) and
590 $\|\nabla_{\theta} r(s, a)\| \leq C$ (Lemma 4), we can easily see that $\|g_{\text{dpo}}(\theta)\| \leq 2TC + \frac{4\log(1+e^{2T\bar{C}})T^2C}{\beta}$.

591 Therefore, we have that $\|g_{\text{ce}}(\theta) + \lambda g_{\text{dpo}}(\theta)\| \leq (1 + 2\lambda)TC + \frac{[2\log(1+e^{T\bar{C}}) + 4\lambda\log(1+e^{2T\bar{C}})]T^2C}{\beta}$.

592 Similarly, we can get that $\|\nabla L(\theta, \pi_{r_{\theta}})\| \leq (1 + 2\lambda)TC + \frac{[2\log(1+e^{T\bar{C}}) + 4\lambda\log(1+e^{2T\bar{C}})]T^2C}{\beta}$. $\square$

593 **Lemma 8.** Suppose Assumption 1 holds, then $\|\nabla_{\theta} \log \pi_{r_{\theta}}(y|x)\| \leq \frac{2CT^2}{\beta}$ and
594 $\|\nabla_{\theta\theta}^2 \log \pi_{r_{\theta}}(y|x)\| \leq \frac{2CT^2 + 4C^2T^4}{\beta}$.

Proof. From Claim 1, we know that

$$\nabla_{\theta} \log \pi_{r_{\theta}}(a_t|s_t) = \nabla_{\theta} \log Z_{r_{\theta}}(s_t, a_t) - \nabla_{\theta} \log Z_{r_{\theta}}(s_t),$$

595 where the terms $\nabla_{\theta} \log Z_{r_{\theta}}(s_t, a_t) = \frac{1}{\beta} E^{\pi_{r_{\theta}}} [\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_t = s_t, A_t = a_t]$ and
596 $\nabla_{\theta} \log Z_{r_{\theta}}(s_t) = \frac{1}{\beta} E^{\pi_{r_{\theta}}} [\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_t = s_t]$. Note that $r_{\theta}(s_t, a_t) = 0$ for $t > T$,
597 thus we can equivalently write $\nabla_{\theta} \log Z_{r_{\theta}}(s_t, a_t) = \frac{1}{\beta} E^{\pi_{r_{\theta}}} [\sum_{i=t}^T \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_t = s_t, A_t =$
598 $a_t]$ and $\nabla_{\theta} \log Z_{r_{\theta}}(s_t) = \frac{1}{\beta} E^{\pi_{r_{\theta}}} [\sum_{i=t}^T \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_t = s_t]$. Therefore, we have that

$$\|\nabla_{\theta} \log \pi_{r_{\theta}}(y|x)\| \leq \sum_{t=1}^{\infty} \|\nabla_{\theta} \log Z_{r_{\theta}}(s_t, a_t)\| + \|\nabla_{\theta} \log Z_{r_{\theta}}(s_t)\| \leq \frac{2CT^2}{\beta}.$$

599 Next, we consider the Hessian:

$$\begin{aligned}
\nabla_{\theta\theta}^2 \log Z_{r_{\theta}}(s, a) &= \frac{1}{\beta} \nabla_{\theta} E^{\pi_{r_{\theta}}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_t = s, A_t = a \right] \\
&= \frac{1}{\beta} \nabla_{\theta\theta}^2 r_{\theta}(s, a) + \frac{1}{\beta} \nabla_{\theta} E^{\pi_{r_{\theta}}} \left[\sum_{i=t+1}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_{t+1} = s' \right] \\
&= \frac{1}{\beta} \nabla_{\theta\theta}^2 r_{\theta}(s, a) + \frac{1}{\beta} \nabla_{\theta} \sum_{a' \in \mathcal{A}} \pi_{r_{\theta}}(a'|s') E^{\pi_{r_{\theta}}} \left[\sum_{i=t+1}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_{t+1} = s', A_{t+1} = a' \right] \\
&= \frac{1}{\beta} \nabla_{\theta\theta}^2 r_{\theta}(s, a) + \frac{1}{\beta} \sum_{a' \in \mathcal{A}} \nabla_{\theta} \pi_{r_{\theta}}(a'|s') \cdot E^{\pi_{r_{\theta}}} \left[\sum_{i=t+1}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_{t+1} = s', A_{t+1} = a' \right] \\
&\quad + \frac{1}{\beta} \sum_{a' \in \mathcal{A}} \pi_{r_{\theta}}(a'|s') \cdot \nabla_{\theta} E^{\pi_{r_{\theta}}} \left[\sum_{i=t+1}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_{t+1} = s', A_{t+1} = a' \right].
\end{aligned}$$

600 Keep expanding $\nabla_{\theta} E^{\pi_{r_{\theta}}} \left[\sum_{i=t+1}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_{t+1} = s', A_{t+1} = a' \right]$, we can get

$$\nabla_{\theta\theta}^2 \log Z_{r_{\theta}}(s, a) = \frac{1}{\beta} E^{\pi_{r_{\theta}}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta\theta}^2 r_{\theta}(S_i, A_i) | S_t = s, A_t = a \right]$$

$$+ \frac{1}{\beta} E^{\pi_{r_\theta}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} \pi_{r_\theta}(A_i | S_i) \cdot E^{\pi_{r_\theta}} \left[\sum_{j=i}^{\infty} \gamma^{j-t} \nabla_{\theta} r_{\theta}(S_j, A_j) \right] \middle| S_t = s, A_t = a \right].$$

601 Now we take a look at the second term in the above equation:

$$\begin{aligned} & \nabla_{\theta} \pi_{r_\theta}(A_i | S_i) \cdot E^{\pi_{r_\theta}} \left[\sum_{j=i}^{\infty} \gamma^{j-t} \nabla_{\theta} r_{\theta}(S_j, A_j) \right] \\ &= \pi_{r_\theta}(A_i | S_i) \nabla_{\theta} \log \pi_{r_\theta}(A_i | S_i) \cdot E^{\pi_{r_\theta}} \left[\sum_{j=i}^{\infty} \gamma^{j-t} \nabla_{\theta} r_{\theta}(S_j, A_j) \right] \\ &\stackrel{(a)}{=} \pi_{r_\theta}(A_i | S_i) \left[\nabla_{\theta} \log Z_{r_\theta}(S_i, A_i) - \nabla_{\theta} \log Z_{r_\theta, i}(S_i) \right] E^{\pi_{r_\theta}} \left[\sum_{j=i}^T \gamma^{j-t} \nabla_{\theta} r_{\theta}(S_j, A_j) \right] \\ &\Rightarrow \left\| \nabla_{\theta} \pi_{r_\theta}(A_i | S_i) \cdot E^{\pi_{r_\theta}} \left[\sum_{j=i}^{\infty} \gamma^{j-t} \nabla_{\theta} r_{\theta}(S_j, A_j) \right] \right\| \leq 2CT \cdot CT = 2C^2T^2. \end{aligned} \quad (13)$$

602 where (a) follows from Claim 1 and that $r_{\theta}(s_t, a_t) \equiv 0$ for $t > T$. Therefore, we have that

$$\begin{aligned} & \|\nabla_{\theta\theta}^2 \log Z_{r_\theta}(s, a)\| \leq \frac{1}{\beta} \left\| E^{\pi_{r_\theta}} \left[\sum_{i=t}^T \gamma^{i-t} \nabla_{\theta\theta}^2 r_{\theta}(S_i, A_i) \middle| S_t = s, A_t = a \right] \right\| \\ &+ \frac{1}{\beta} \left\| E^{\pi_{r_\theta}} \left[\sum_{i=t}^T \gamma^{i-t} \nabla_{\theta} \pi_{r_\theta}(A_i | S_i) \cdot E^{\pi_{r_\theta}} \left[\sum_{j=i}^T \gamma^{j-t} \nabla_{\theta} r_{\theta}(S_j, A_j) \right] \middle| S_t = s, A_t = a \right] \right\| \\ &\stackrel{(b)}{\leq} \frac{\bar{C}T}{\beta} + \frac{1}{\beta} \cdot T \cdot 2C^2T^2 = \frac{\bar{C}T + 2C^2T^3}{\beta}. \end{aligned}$$

603 where (b) follows from Lemma 4 and (13). Similarly, we can get $\|\nabla_{\theta\theta}^2 \log Z_{r_\theta}(s)\| \leq \frac{\bar{C}T + 2C^2T^3}{\beta}$.

604 Therefore, we have that

$$\begin{aligned} \|\nabla_{\theta\theta}^2 \log \pi_{r_\theta}(y|x)\| &\leq \sum_{t=1}^{\infty} \|\nabla_{\theta\theta}^2 \log \pi_{r_\theta}(a_t | s_t)\| \\ &\leq \sum_{t=1}^{\infty} \left[\|\nabla_{\theta\theta}^2 \log Z_{r_\theta}(s, a)\| + \|\nabla_{\theta\theta}^2 \log Z_{r_\theta}(s)\| \right] \\ &\leq \frac{2\bar{C}T^2 + 4C^2T^4}{\beta}. \end{aligned}$$

605

□

606 **Lemma 9.** Suppose Assumption 1 holds, then $\|\nabla^2 L(\theta, \pi_{r_\theta})\| \leq C_L$ where C_L is a positive constant
607 whose expression is in (16).

608 *Proof.* Since $L(\theta, \pi_{r_\theta}) = L_{\text{ce}}(\theta, \pi_{r_\theta}) + \lambda L_{\text{dpo}}(\theta, \pi_{r_\theta})$, we compute the Hessian of $L_{\text{ce}}(\theta, \pi_{r_\theta})$ and
609 $L_{\text{dpo}}(\theta, \pi_{r_\theta})$ respectively.

610 Recall from (5) that

$$\nabla_{\theta} L_{\text{ce}}(\theta, \pi_{r_\theta}) = E_{x \sim \mathcal{D}, y \sim \pi_{r_\theta}(\cdot|x)} \left[l_{\text{ce}}(\theta; (x, y)) \nabla_{\theta} \log \pi_{r_\theta}(y|x) + \nabla_{\theta} l_{\text{ce}}(\theta; (x, y)) \right].$$

611 Define

$$\phi(\theta; (x, y)) \triangleq l_{\text{ce}}(\theta; (x, y)) \nabla_{\theta} \log \pi_{r_\theta}(y|x) + \nabla_{\theta} l_{\text{ce}}(\theta; (x, y)).$$

612 Then

$$\nabla_{\theta\theta}^2 L_{\text{ce}}(\theta, \pi_{r_\theta}) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{r_\theta}(\cdot|x)} \left[\phi(\theta; (x, y)) \nabla_{\theta} \log \pi_{r_\theta}(y|x)^{\top} + \nabla_{\theta} \phi(\theta; (x, y)) \right].$$

613 Expanding the first term,

$$\begin{aligned} \phi(\theta; (x, y)) \nabla_{\theta} \log \pi_{r_{\theta}}(y|x)^{\top} &= l_{\text{ce}}(\theta; (x, y)) \nabla_{\theta} \log \pi_{r_{\theta}}(y|x) \nabla_{\theta} \log \pi_{r_{\theta}}(y|x)^{\top} \\ &\quad + \nabla_{\theta} l_{\text{ce}}(\theta; (x, y)) \nabla_{\theta} \log \pi_{r_{\theta}}(y|x)^{\top}. \end{aligned}$$

614 Moreover,

$$\begin{aligned} \nabla_{\theta} \phi(\theta; (x, y)) &= \nabla_{\theta} \left(l_{\text{ce}}(\theta; (x, y)) \nabla_{\theta} \log \pi_{r_{\theta}}(y|x) + \nabla_{\theta} l_{\text{ce}}(\theta; (x, y)) \right) \\ &= \nabla_{\theta} \log \pi_{r_{\theta}}(y|x) \nabla_{\theta} l_{\text{ce}}(\theta; (x, y))^{\top} + l_{\text{ce}}(\theta; (x, y)) \nabla_{\theta\theta}^2 \log \pi_{r_{\theta}}(y|x) + \nabla_{\theta\theta}^2 l_{\text{ce}}(\theta; (x, y)). \end{aligned}$$

615 Therefore,

$$\begin{aligned} \nabla_{\theta\theta}^2 L_{\text{ce}}(\theta, \pi_{r_{\theta}}) &= \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{r_{\theta}}(\cdot|x)} \left[l_{\text{ce}}(\theta; (x, y)) \nabla_{\theta} \log \pi_{r_{\theta}}(y|x) \nabla_{\theta} \log \pi_{r_{\theta}}(y|x)^{\top} \right. \\ &\quad \left. + \nabla_{\theta} l_{\text{ce}}(\theta; (x, y)) \nabla_{\theta} \log \pi_{r_{\theta}}(y|x)^{\top} + \nabla_{\theta} \log \pi_{r_{\theta}}(y|x) \nabla_{\theta} l_{\text{ce}}(\theta; (x, y))^{\top} \right. \\ &\quad \left. + l_{\text{ce}}(\theta; (x, y)) \nabla_{\theta\theta}^2 \log \pi_{r_{\theta}}(y|x) + \nabla_{\theta\theta}^2 l_{\text{ce}}(\theta; (x, y)) \right]. \end{aligned}$$

616 Since $|l_{\text{ce}}(\theta; (x, y))| \leq \log(1 + e^{T\bar{C}})$, $\|\nabla_{\theta} l_{\text{ce}}(\theta; (x, y))\| \leq TC$, $\|\nabla_{\theta\theta}^2 l_{\text{ce}}(\theta; (x, y))\| \leq \frac{1}{4}T^2C^2 +$
 617 $T\bar{C}$ (Lemma 5), and $\|\nabla_{\theta} \log \pi_{r_{\theta}}(y|x)\| \leq \frac{2CT^2}{\beta}$, $\|\nabla_{\theta\theta}^2 \log \pi_{r_{\theta}}(y|x)\| \leq \frac{2\bar{C}T^2 + 4C^2T^4}{\beta}$ (Lemma 8).

618 Therefore, we can see that

$$\begin{aligned} &\|\nabla_{\theta\theta}^2 L_{\text{ce}}(\theta, \pi_{r_{\theta}})\| \\ &\leq \frac{4 \log(1 + e^{T\bar{C}}) C^2 T^4}{\beta^2} + \frac{4C^2 T^3}{\beta} + \frac{(2\bar{C}T^2 + 4C^2T^4) \log(1 + e^{T\bar{C}})}{\beta} + \frac{1}{4}T^2C^2 + T\bar{C}. \quad (14) \end{aligned}$$

619 Next, we compute the Hessian of $L_{\text{dpo}}(\theta, \pi_{r_{\theta}})$. Recall from (9) that

$$\begin{aligned} \nabla_{\theta} L_{\text{dpo}}(\theta, \pi_{r_{\theta}}) &= \mathbb{E}_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_{r_{\theta}}(\cdot, \cdot|x)} \left[l_{\text{dpo}}(\theta; (x, y^+, y^-)) \nabla_{\theta} \log \pi_{r_{\theta}}(y^+, y^-|x) + \nabla_{\theta} l_{\text{dpo}}(\theta; (x, y^+, y^-)) \right]. \end{aligned}$$

620 Define

$$\phi(\theta; (x, y^+, y^-)) \triangleq l_{\text{dpo}}(\theta; (x, y^+, y^-)) \nabla_{\theta} \log \pi_{r_{\theta}}(y^+, y^-|x) + \nabla_{\theta} l_{\text{dpo}}(\theta; (x, y^+, y^-)).$$

621 Then

$$\begin{aligned} \nabla_{\theta\theta}^2 L_{\text{dpo}}(\theta, \pi_{r_{\theta}}) &= \mathbb{E}_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_{r_{\theta}}(\cdot, \cdot|x)} \left[\phi(\theta; (x, y^+, y^-)) \nabla_{\theta} \log \pi_{r_{\theta}}(y^+, y^-|x)^{\top} + \nabla_{\theta} \phi(\theta; (x, y^+, y^-)) \right]. \end{aligned}$$

622 Expanding the first term,

$$\begin{aligned} &\phi(\theta; (x, y^+, y^-)) \nabla_{\theta} \log \pi_{r_{\theta}}(y^+, y^-|x)^{\top} \\ &= l_{\text{dpo}}(\theta; (x, y^+, y^-)) \nabla_{\theta} \log \pi_{r_{\theta}}(y^+, y^-|x) \nabla_{\theta} \log \pi_{r_{\theta}}(y^+, y^-|x)^{\top} \\ &\quad + \nabla_{\theta} l_{\text{dpo}}(\theta; (x, y^+, y^-)) \nabla_{\theta} \log \pi_{r_{\theta}}(y^+, y^-|x)^{\top}. \end{aligned}$$

623 Moreover,

$$\begin{aligned} \nabla_{\theta} \phi(\theta; (x, y^+, y^-)) &= \nabla_{\theta} \left(l_{\text{dpo}}(\theta; (x, y^+, y^-)) \nabla_{\theta} \log \pi_{r_{\theta}}(y^+, y^-|x) + \nabla_{\theta} l_{\text{dpo}}(\theta; (x, y^+, y^-)) \right) \\ &= \nabla_{\theta} \log \pi_{r_{\theta}}(y^+, y^-|x) \nabla_{\theta} l_{\text{dpo}}(\theta; (x, y^+, y^-))^{\top} \\ &\quad + l_{\text{dpo}}(\theta; (x, y^+, y^-)) \nabla_{\theta\theta}^2 \log \pi_{r_{\theta}}(y^+, y^-|x) + \nabla_{\theta\theta}^2 l_{\text{dpo}}(\theta; (x, y^+, y^-)). \end{aligned}$$

624 Therefore,

$$\begin{aligned} &\nabla_{\theta\theta}^2 L_{\text{dpo}}(\theta, \pi_{r_{\theta}}) \\ &= \mathbb{E}_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_{r_{\theta}}(\cdot, \cdot|x)} \left[l_{\text{dpo}}(\theta; (x, y^+, y^-)) \nabla_{\theta} \log \pi_{r_{\theta}}(y^+, y^-|x) \nabla_{\theta} \log \pi_{r_{\theta}}(y^+, y^-|x)^{\top} \right. \\ &\quad \left. + \nabla_{\theta} l_{\text{dpo}}(\theta; (x, y^+, y^-)) \nabla_{\theta} \log \pi_{r_{\theta}}(y^+, y^-|x)^{\top} \right. \end{aligned}$$

$$+ \nabla_{\theta} \log \pi_{r_{\theta}}(y^+, y^- | x) \nabla_{\theta} l_{\text{dpo}}(\theta; (x, y^+, y^-))^{\top} \\ + l_{\text{dpo}}(\theta; (x, y^+, y^-)) \nabla_{\theta\theta}^2 \log \pi_{r_{\theta}}(y^+, y^- | x) + \nabla_{\theta\theta}^2 l_{\text{dpo}}(\theta; (x, y^+, y^-)) \Big].$$

625 Since $\log \pi_{r_{\theta}}(y^+, y^- | x) = \log \pi_{r_{\theta}}(y^+ | x) + \log \pi_{r_{\theta}}(y^- | x)$, then from Lemma 8, we can see that
 626 $\|\nabla_{\theta} \log \pi_{r_{\theta}}(y^+, y^- | x)\| \leq \frac{4CT^2}{\beta}$ and $\|\nabla_{\theta\theta}^2 \log \pi_{r_{\theta}}(y^+, y^- | x)\| \leq \frac{4\bar{C}T^2 + 8C^2T^4}{\beta}$. From Lemma
 627 6, we know that $|l_{\text{dpo}}(\theta; (x, y^+, y^-))| \leq \log(1 + e^{2T\bar{C}})$, $\|\nabla_{\theta} l_{\text{dpo}}(\theta; (x, y^+, y^-))\| \leq 2TC$, and
 628 $\|\nabla_{\theta\theta}^2 l_{\text{dpo}}(\theta; (x, y^+, y^-))\| \leq T^2C^2 + 2T\bar{C}$. Therefore, we can bound the Hessian:

$$\|\nabla_{\theta\theta}^2 L_{\text{dpo}}(\theta, \pi_{r_{\theta}})\| \tag{15} \\ \leq \frac{16 \log(1 + e^{2T\bar{C}}) C^2 T^4}{\beta^2} + \frac{16C^2 T^3}{\beta} + \frac{(4\bar{C}T^2 + 8C^2 T^4) \log(1 + e^{2T\bar{C}})}{\beta} + T^2 C^2 + 2T\bar{C}.$$

Combining (14) and (15), we can reach that

$$\|\nabla^2 L(\theta, \pi_{r_{\theta}})\| \leq C_L,$$

629 where

$$C_L = \frac{4 \log(1 + e^{T\bar{C}}) C^2 T^4}{\beta^2} + \frac{4C^2 T^3}{\beta} + \frac{(2\bar{C}T^2 + 4C^2 T^4) \log(1 + e^{T\bar{C}})}{\beta} + \frac{1}{4} T^2 C^2 + T\bar{C} \\ \lambda \left[\frac{16 \log(1 + e^{2T\bar{C}}) C^2 T^4}{\beta^2} + \frac{16C^2 T^3}{\beta} + \frac{(4\bar{C}T^2 + 8C^2 T^4) \log(1 + e^{2T\bar{C}})}{\beta} + T^2 C^2 + 2T\bar{C} \right]. \tag{16}$$

630 $\square$

631 B.4 Proof of Theorem 1 (lower-level convergence)

632 Theorem 1 includes both lower-level convergence and upper-level convergence. We start with
 633 lower-level convergence:

634 We use soft policy iteration as the lower-level policy update for analysis. At each iteration k , we have
 635 a policy π_k and a reward function r_{θ_k} , a one-step policy update to get π_{k+1} as follows from:

$$\pi_{k+1}(a|s) = \frac{\pi_{\text{ref}}(a|s) \exp(\frac{1}{\beta} Q_{r_{\theta_k}}^{\pi_k}(s, a))}{\sum_{\bar{a} \in \mathcal{A}} \pi_{\text{ref}}(\bar{a}|s) \exp(\frac{1}{\beta} Q_{r_{\theta_k}}^{\pi_k}(s, \bar{a}))} \\ Q_{r_{\theta_k}}^{\pi_k}(s, a) = r_{\theta_k}(s, a) + \gamma V_{r_{\theta_k}}^{\pi_k}(s') \\ V_{r_{\theta_k}}^{\pi_k}(s) = \beta \log \left[\sum_{\bar{a} \in \mathcal{A}} \pi_{\text{ref}}(\bar{a}|s) \exp(\frac{1}{\beta} Q_{r_{\theta_k}}^{\pi_k}(s, \bar{a})) \right],$$

636 where $s' = (s, a)$.

637 Define the soft Bellman operator

$$(\mathcal{T}_{\theta_k} Q)(s, a) \triangleq r_{\theta_k}(s, a) + \gamma \beta \log \left[\sum_{\bar{a}' \in \mathcal{A}} \pi_{\text{ref}}(\bar{a}'|s') \exp(\frac{1}{\beta} Q(s', \bar{a}')) \right].$$

638 Then one-step soft policy iteration is equivalently written as

$$Q_{k+1} = \mathcal{T}_{\theta_k}(Q_k), \quad \pi_k(s, a) \propto \pi_{\text{ref}}(a|s) \exp(\frac{1}{\beta} Q_k(s, a)).$$

Claim 2. For two action-value functions Q and Q' where $Q, Q' \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$, it holds that

$$\|\mathcal{T}_{\theta_k}(Q) - \mathcal{T}_{\theta_k}(Q')\|_{\infty} \leq \gamma \|Q - Q'\|_{\infty}.$$

639 *Proof.* We define $\|Q - Q'\|_{\infty} \triangleq \max_{s \in \mathcal{S}, a \in \mathcal{A}} |Q(s, a) - Q'(s, a)|$ and denote $\epsilon = \|Q - Q'\|_{\infty}$.
 640 Therefore, we have that

$$\gamma \beta \log \left[\sum_{a \in \mathcal{A}} \pi_{\text{ref}}(a|s) \exp(\frac{1}{\beta} Q(s, a)) \right] \leq \gamma \beta \log \left[\sum_{a \in \mathcal{A}} \pi_{\text{ref}}(a|s) \exp(\frac{1}{\beta} (Q'(s, a) + \epsilon)) \right]$$

$$= \gamma\epsilon + \gamma\beta \log \left[\sum_{a \in \mathcal{A}} \pi_{\text{ref}}(a|s) \exp\left(\frac{1}{\beta} Q'(s, a)\right) \right].$$

Similarly, we can get that

$$\gamma\beta \log \left[\sum_{a \in \mathcal{A}} \pi_{\text{ref}}(a|s) \exp\left(\frac{1}{\beta} Q(s, a)\right) \right] \geq -\gamma\epsilon + \gamma\beta \log \left[\sum_{a \in \mathcal{A}} \pi_{\text{ref}}(a|s) \exp\left(\frac{1}{\beta} Q'(s, a)\right) \right].$$

Therefore, $\|\mathcal{T}_{\theta_k}(Q) - \mathcal{T}_{\theta_k}(Q')\|_{\infty} \leq \gamma\epsilon = \gamma\|Q - Q'\|_{\infty}$. $\square$

Claim 3. It holds that $Q_{r_{\theta_k}}^{\pi_{k+1}}(s, a) \geq \mathcal{T}_{\theta_k}(Q_{r_{\theta_k}}^{\pi_k}(s, a))$ for any (s, a) .

Proof.

$$\begin{aligned} Q_{r_{\theta_k}}^{\pi_{k+1}}(s, a) &\stackrel{(a)}{=} r_{\theta_k}(s, a) + \gamma \sum_{a' \in \mathcal{A}} \pi_{k+1}(a'|s') \left[Q_{r_{\theta_k}}^{\pi_{k+1}}(s', a') - \beta \log \frac{\pi_{k+1}(a'|s')}{\pi_{\text{ref}}(a'|s')} \right] \\ &\stackrel{(b)}{\geq} r_{\theta_k}(s, a) + \gamma \sum_{a' \in \mathcal{A}} \pi_{k+1}(a'|s') \left[Q_{r_{\theta_k}}^{\pi_k}(s', a') - \beta \log \frac{\pi_{k+1}(a'|s')}{\pi_{\text{ref}}(a'|s')} \right] \\ &= r_{\theta_k}(s, a) + \gamma\beta \log \left[\sum_{a' \in \mathcal{A}} \pi_{\text{ref}}(a'|s') \exp\left(\frac{1}{\beta} Q_{r_{\theta_k}}^{\pi_k}(s', a')\right) \right] \\ &= \mathcal{T}_{\theta_k}(Q_{r_{\theta_k}}^{\pi_k}(s, a)), \end{aligned}$$

where (a) follows from equations (2)-(3) in [39] and (b) follows from policy improvement theorem (Theorem 4 in [29]). $\square$

Claim 4. Suppose Assumption 1 holds. For an arbitrary policy π and reward parameters θ_1 and θ_2 , it holds that

$$\begin{aligned} |Q_{r_{\theta_1}}^*(s, a) - Q_{r_{\theta_2}}^*(s, a)| &\leq \hat{C} \|\theta_1 - \theta_2\| \\ |Q_{r_{\theta_1}}^{\pi}(s, a) - Q_{r_{\theta_2}}^{\pi}(s, a)| &\leq \hat{C} \|\theta_1 - \theta_2\|, \end{aligned}$$

for any $(s, a) \in \mathcal{S} \times \mathcal{A}$, and $\hat{C} = CT$ is a positive constant.

Proof. Recall from the proof of Claim 1 that $Z_{r_{\theta}}(s, a) \triangleq \pi_{\text{ref}}(a|s) \exp(\frac{1}{\beta} Q_{r_{\theta}}^*(s, a))$ and $\nabla_{\theta} \log Z_{r_{\theta}}(s, a) = \frac{1}{\beta} E^{\pi_{r_{\theta_k}}} [\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_t = s, A_t = a]$. Therefore, we have that

$$\begin{aligned} \nabla_{\theta} Q_{r_{\theta}}^{\pi}(s, a) &= \beta \nabla_{\theta} \log Z_{r_{\theta}}^{\pi}(s, a) \\ &= E^{\pi} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_t = s, A_t = a \right] \\ &\stackrel{(c)}{\leq} CT, \end{aligned}$$

where (c) follows from Lemma 4 that $\|\nabla_{\theta} r_{\theta}(s, a)\| \leq C$ and that $r_{\theta}(s_t, a_t) \equiv 0$ for $t > T$.

The action value function can be equivalently reformulated as [29]:

$$Q_r^{\pi}(s, a) = E^{\pi} \left[\sum_{i=t}^{\infty} \gamma^{i-t} r(S_i, A_i) - \beta \log \frac{\pi(A_i|S_i)}{\pi_{\text{ref}}(A_i|S_i)} \middle| S_t = s, A_t = a \right]. \quad (17)$$

Therefore, we have that

$$\begin{aligned} &\left| Q_{r_{\theta_1}}^{\pi}(s, a) - Q_{r_{\theta_2}}^{\pi}(s, a) \right| \\ &\stackrel{(d)}{=} \left| E^{\pi} \left[\sum_{i=t}^{\infty} \gamma^{i-t} r_{\theta_1}(S_i, A_i) - \beta \log \frac{\pi(A_i|S_i)}{\pi_{\text{ref}}(A_i|S_i)} \middle| S_t = s, A_t = a \right] \right. \\ &\quad \left. - E^{\pi} \left[\sum_{i=t}^{\infty} \gamma^{i-t} r_{\theta_2}(S_i, A_i) - \beta \log \frac{\pi(A_i|S_i)}{\pi_{\text{ref}}(A_i|S_i)} \middle| S_t = s, A_t = a \right] \right| \end{aligned}$$

$$\begin{aligned}
&= \left| E^\pi \left[\sum_{i=t}^{\infty} \gamma^{i-t} r_{\theta_1}(S_i, A_i) - r_{\theta_2}(S_i, A_i) \middle| S_t = s, A_t = a \right] \right| \\
&\stackrel{(e)}{\leq} CT \|\theta_1 - \theta_2\|,
\end{aligned}$$

653 where (d) follows from (17) and (e) follows from Lemma 4. Note that on a convex domain,
654 $|f(x_1) - f(x_2)| \leq L\|x_1 - x_2\|$ and $\|\nabla f(x)\| \leq L$ are equivalent [40, 41], where L is a positive
655 constant. $\square$

656 Now, we proceed to quantify the convergence of the lower-level policy in Algorithm 1. For any
657 $(s, a) \in \mathcal{S} \times \mathcal{A}$, we know that

$$\begin{aligned}
&|\log \pi_{k+1}(a|s) - \log \pi_{\theta_k}(a|s)| \\
&\leq \frac{1}{\beta} (|Q_{r_{\theta_k}}^{\pi_k}(s, a) - Q_{r_{\theta_k}}^*(s, a)| + |V_{r_{\theta_k}}^{\pi_k}(s) - V_{r_{\theta_k}}^*(s)|) \\
&\stackrel{(f)}{\leq} \frac{1}{\beta} (|Q_{r_{\theta_k}}^{\pi_k}(s, a) - Q_{r_{\theta_k}}^*(s, a)| + \max_{(s,a) \in \mathcal{S} \times \mathcal{A}} \{|Q_{r_{\theta_k}}^{\pi_k}(s, a) - Q_{r_{\theta_k}}^*(s, a)|\}) \\
&\leq \frac{2}{\beta} \|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_{\infty}, \tag{18}
\end{aligned}$$

658 where (f) follows from equation (48) in [42].

659 Now we take a look at the term $\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_{\infty}$:

$$\begin{aligned}
\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_{\infty} &\leq \|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_{k-1}}}^{\pi_k}\|_{\infty} + \|Q_{r_{\theta_{k-1}}}^{\pi_k} - Q_{r_{\theta_{k-1}}}^*\|_{\infty} + \|Q_{r_{\theta_{k-1}}}^* - Q_{r_{\theta_k}}^*\|_{\infty} \\
&\stackrel{(g)}{\leq} \hat{C} \|\theta_{k-1} - \theta_k\| + \|Q_{r_{\theta_{k-1}}}^{\pi_k} - Q_{r_{\theta_{k-1}}}^*\|_{\infty} + \hat{C} \|\theta_{k-1} - \theta_k\| \\
&\stackrel{(h)}{=} 2\hat{C} \|\theta_{k-1} - \theta_k\| + \|Q_{r_{\theta_{k-1}}}^* - Q_{r_{\theta_{k-1}}}^{\pi_{k-1}}\|_{\infty} \\
&\stackrel{(i)}{\leq} 2\hat{C} \|\theta_{k-1} - \theta_k\| + \|Q_{r_{\theta_{k-1}}}^* - \mathcal{T}_{\theta_{k-1}}(Q_{r_{\theta_{k-1}}}^{\pi_{k-1}})\|_{\infty} \\
&\stackrel{(j)}{=} 2\hat{C} \|\theta_{k-1} - \theta_k\| + \|\mathcal{T}_{\theta_{k-1}}(Q_{r_{\theta_{k-1}}}^*) - \mathcal{T}_{\theta_{k-1}}(Q_{r_{\theta_{k-1}}}^{\pi_{k-1}})\|_{\infty} \\
&\stackrel{(k)}{\leq} 2\hat{C} \|\theta_{k-1} - \theta_k\| + \gamma \|Q_{r_{\theta_{k-1}}}^* - Q_{r_{\theta_{k-1}}}^{\pi_{k-1}}\|_{\infty}, \tag{19}
\end{aligned}$$

660 where (g) follows from Claim 4, (h) follows from the fact that $Q_{r_{\theta_{k-1}}}^*(s, a) \geq Q_{r_{\theta_{k-1}}}^{\pi_{k-1}}(s, a)$ for all
661 $(s, a) \in \mathcal{S} \times \mathcal{A}$ because $Q_{r_{\theta_{k-1}}}^*$ is the optimal action-value function under $r_{\theta_{k-1}}$, (i) follows from
662 Claim 3, (j) follows from the fact that $Q_{r_{\theta_{k-1}}}^* = \mathcal{T}_{\theta_{k-1}}(Q_{r_{\theta_{k-1}}}^*)$ because $Q_{r_{\theta_{k-1}}}^*$ is the fixed point
663 of the contraction map $\mathcal{T}_{\theta_{k-1}}$ [29], and (k) follows from Claim 2. Therefore, we have that

$$\begin{aligned}
&\sum_{k=1}^{K-1} \|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_{\infty} \\
&\leq 2\hat{C} \sum_{k=1}^{K-1} \|\theta_{k-1} - \theta_k\| + \gamma \sum_{k=1}^{K-1} \|Q_{r_{\theta_{k-1}}}^* - Q_{r_{\theta_{k-1}}}^{\pi_{k-1}}\|_{\infty} \\
&= 2\hat{C} \sum_{k=1}^{K-1} \alpha \|g_{\text{ce}}(\theta_{k-1}) + \lambda g_{\text{dpo}}(\theta_{k-1})\|_{\infty} + \gamma \sum_{k=1}^{K-1} \|Q_{r_{\theta_{k-1}}}^* - Q_{r_{\theta_{k-1}}}^{\pi_{k-1}}\|_{\infty} \\
&\stackrel{(l)}{\leq} 2\hat{C} \sum_{k=1}^{K-1} \alpha \left[(1 + 2\lambda)TC + \frac{[2\log(1 + e^{T\hat{C}}) + 4\lambda\log(1 + e^{2T\hat{C}})]T^2C}{\beta} \right] \\
&\quad + \gamma \sum_{k=1}^{K-1} \|Q_{r_{\theta_{k-1}}}^* - Q_{r_{\theta_{k-1}}}^{\pi_{k-1}}\|_{\infty},
\end{aligned}$$

664 where (l) follows from Lemma 7. By subtracting $\gamma \sum_{k=1}^{K-1} \|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_\infty$ on both sides, we have
 665 that

$$\begin{aligned} & (1 - \gamma) \sum_{k=1}^{K-1} \|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_\infty \\ & \leq 2\hat{C} \left[(1 + 2\lambda)TC + \frac{[2\log(1 + e^{T\tilde{C}}) + 4\lambda\log(1 + e^{2T\tilde{C}})]T^2C}{\beta} \right] \sum_{k=1}^K \alpha \\ & \quad + \gamma \left(\|Q_{r_{\theta_0}}^* - Q_{r_{\theta_0}}^{\pi_0}\|_\infty - \|Q_{r_{\theta_{K-1}}}^* - Q_{r_{\theta_{K-1}}}^{\pi_{K-1}}\|_\infty \right). \end{aligned}$$

666 Dividing both sides by $(1 - \gamma)$, we have that

$$\begin{aligned} & \sum_{k=0}^{K-1} \|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_\infty = \|Q_{r_{\theta_0}}^{\pi_0} - Q_{r_{\theta_0}}^*\|_\infty + \sum_{k=1}^{K-1} \|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_\infty \\ & \leq \frac{2\hat{C}}{1 - \gamma} \left[(1 + 2\lambda)TC + \frac{[2\log(1 + e^{T\tilde{C}}) + 4\lambda\log(1 + e^{2T\tilde{C}})]T^2C}{\beta} \right] \sum_{k=1}^K \alpha \\ & \quad + \frac{\gamma}{(1 - \gamma)} \left(\|Q_{r_{\theta_0}}^* - Q_{r_{\theta_0}}^{\pi_0}\|_\infty - \|Q_{r_{\theta_K}}^* - Q_{r_{\theta_K}}^{\pi_K}\|_\infty \right) + \|Q_{r_{\theta_0}}^{\pi_0} - Q_{r_{\theta_0}}^*\|_\infty. \end{aligned} \quad (20)$$

667 Finally, we have that

$$\begin{aligned} & \frac{1}{K} \sum_{k=0}^{K-1} |\log \pi_{k+1} - \log \pi_{\theta_k}|_\infty \\ & \stackrel{(m)}{\leq} \frac{2}{\beta K} \sum_{k=0}^{K-1} \|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_\infty \\ & \stackrel{(n)}{\leq} \frac{4\hat{C}}{(1 - \gamma)\beta K} \left[(1 + 2\lambda)TC + \frac{[2\log(1 + e^{T\tilde{C}}) + 4\lambda\log(1 + e^{2T\tilde{C}})]T^2C}{\beta} \right] \sum_{k=1}^K K^{-1/2} \\ & \quad + \frac{2\gamma}{(1 - \gamma)\beta K} \left(\|Q_{r_{\theta_0}}^* - Q_{r_{\theta_0}}^{\pi_0}\|_\infty - \|Q_{r_{\theta_K}}^* - Q_{r_{\theta_K}}^{\pi_K}\|_\infty \right) + \frac{1}{K} \|Q_{r_{\theta_0}}^{\pi_0} - Q_{r_{\theta_0}}^*\|_\infty \\ & = O(1/\sqrt{K}) + O(1/K), \end{aligned}$$

668 where (m) follows from (18) and (n) follows from (20).

669 B.5 Proof of Theorem 1 (upper-level convergence)

670 We next quantify the convergence of upper level. For simple notations, we use $L(\theta)$ to represent
 671 $L(\theta, \pi_{r_\theta})$ and use $g(\theta)$ to represent $g_{\text{ce}}(\theta) + \lambda g_{\text{dpo}}(\theta)$:

$$\begin{aligned} L(\theta_{k+1}) & \stackrel{(a)}{\leq} L(\theta_k) + \langle \nabla L(\theta_k), \theta_{k+1} - \theta_k \rangle + \frac{C_L}{2} \|\theta_{k+1} - \theta_k\|^2 \\ & = L(\theta_k) - \alpha \langle \nabla L(\theta_k), g(\theta_k) \rangle + \frac{C_L \alpha^2}{2} \|g(\theta_k)\|^2 \\ & = L(\theta_k) - \alpha \langle \nabla L(\theta_k), E[g(\theta_k) - \nabla L(\theta_k)] \rangle - \alpha \|\nabla L(\theta_k)\|^2 + \frac{C_L \alpha^2}{2} \|g(\theta_k)\|^2 \\ & \stackrel{(b)}{\leq} L(\theta_k) - \alpha \langle \nabla L(\theta_k), E[g(\theta_k) - \nabla L(\theta_k)] \rangle - \alpha \|\nabla L(\theta_k)\|^2 + \frac{C_L C_g \alpha^2}{2}. \end{aligned}$$

672 where (a) follows from Lemma 9 and (b) follows from Lemma 7 such that $C_g = (1 + 2\lambda)TC +$
 673 $\frac{[2\log(1 + e^{T\tilde{C}}) + 4\lambda\log(1 + e^{2T\tilde{C}})]T^2C}{\beta}$. Take expectation on both sides, we have that

$$E[L(\theta_{k+1})] \leq E[L(\theta_k)] - \alpha E[\langle \nabla L(\theta_k), E[g(\theta_k) - \nabla L(\theta_k) | \theta_k] \rangle] - \alpha E[\|\nabla L(\theta_k)\|^2] + \frac{C_L C_g \alpha^2}{2}. \quad (21)$$

674 We next start from the following Claim, which is proved in the literature:

675 **Claim 5** (inequality (64) in [42]). *It holds that* $\left\| E_{x \sim \mathcal{D}, y \sim \pi_{r_{\theta_k}}(\cdot|x)} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) \right] - \right.$
676 $E_{x \sim \mathcal{D}, y \sim \pi_k(\cdot|x)} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) \right] \left\| \leq CC_d \sqrt{|\mathcal{S}| \times |\mathcal{A}|} E \left[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^* \|_{\infty} \right]$ *where*
677 $\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^* \|_{\infty} \triangleq \max_{(s,a) \in \mathcal{S} \times \mathcal{A}} \{ |Q_{r_{\theta_k}}^{\pi_k}(s, a) - Q_{r_{\theta_k}}^*(s, a)| \}.$

678 Based on Claim 5, we can see that

$$\begin{aligned}
& \left\| E_{x \sim \mathcal{D}, y \sim \pi_{r_{\theta_k}}(\cdot|x)} \left[\nabla_{\theta} \log \pi_{r_{\theta}}(y|x) \right] - E_{x \sim \mathcal{D}, y \sim \pi_k(\cdot|x)} \left[\nabla_{\theta} \log \pi_{r_{\theta}}(y|x) \right] \right\| \\
& \leq \sum_{t=1}^{\infty} \left\| E_{x \sim \mathcal{D}, y \sim \pi_{r_{\theta_k}}(\cdot|x)} \left[\nabla_{\theta} \log \pi_{r_{\theta}}(a_t|s_t) \right] - E_{x \sim \mathcal{D}, y \sim \pi_k(\cdot|x)} \left[\nabla_{\theta} \log \pi_{r_{\theta}}(a_t|s_t) \right] \right\| \\
& \stackrel{(c)}{=} \sum_{t=1}^{\infty} \left\| E_{x \sim \mathcal{D}, y \sim \pi_{r_{\theta_k}}(\cdot|x)} \left[E^{\pi_{r_{\theta}}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_t = s_t, A_t = a_t \right] \right. \right. \\
& \quad \left. \left. - E^{\pi_{r_{\theta}}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_t = s_t \right] \right] \right. \\
& \quad \left. - E_{x \sim \mathcal{D}, y \sim \pi_k(\cdot|x)} \left[E^{\pi_{r_{\theta}}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_t = s_t, A_t = a_t \right] \right] \right. \\
& \quad \left. - E^{\pi_{r_{\theta}}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_t = s_t \right] \right\| \\
& \leq \sum_{t=1}^{\infty} \left\{ \left\| E_{x \sim \mathcal{D}, y \sim \pi_{r_{\theta_k}}(\cdot|x)} \left[E^{\pi_{r_{\theta}}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_t = s_t, A_t = a_t \right] \right] \right. \right. \\
& \quad \left. \left. - E_{x \sim \mathcal{D}, y \sim \pi_k(\cdot|x)} \left[E^{\pi_{r_{\theta}}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_t = s_t, A_t = a_t \right] \right] \right\| \right. \\
& \quad \left. + \left\| E_{x \sim \mathcal{D}, y \sim \pi_{r_{\theta_k}}(\cdot|x)} \left[E^{\pi_{r_{\theta}}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_t = s_t \right] \right] \right. \right. \\
& \quad \left. \left. - E_{x \sim \mathcal{D}, y \sim \pi_k(\cdot|x)} \left[E^{\pi_{r_{\theta}}} \left[\sum_{i=t}^{\infty} \gamma^{i-t} \nabla_{\theta} r_{\theta}(S_i, A_i) | S_t = s_t \right] \right] \right\| \right\} \\
& \stackrel{(d)}{\leq} 2TCC_d \sqrt{|\mathcal{S}| \times |\mathcal{A}|} E \left[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^* \|_{\infty} \right] \tag{22}
\end{aligned}$$

679 where (c) follows from Claim 1 and (d) follows from Claim 5 and that $r_{\theta}(s_t, a_t) \equiv 0$ for $t > T$.

680 Now, we take a look at the second term in (21)

$$\begin{aligned}
& E[\langle \nabla L(\theta_k), E[g(\theta_k) - \nabla L(\theta_k) | \theta_k] \rangle] \\
& = E[\langle \nabla L(\theta_k), E[(g_{\text{ce}}(\theta_k) - \nabla L_{\text{ce}}(\theta_k)) + \lambda(g_{\text{dpo}}(\theta_k) - \nabla L_{\text{dpo}}(\theta_k)) | \theta_k] \rangle] \\
& \stackrel{(e)}{\leq} C_{\nabla L} E \left[\left\| E[g_{\text{ce}}(\theta_k) - \nabla L_{\text{ce}}(\theta_k)] \right\| + \lambda \left\| E[g_{\text{dpo}}(\theta_k) - \nabla L_{\text{dpo}}(\theta_k)] \right\| \right], \tag{23}
\end{aligned}$$

681 where (e) follows from Lemma 7 such that $C_{\nabla L} = (1 + 2\lambda)TC + \frac{[2\log(1+e^{T\tilde{C}}) + 4\lambda\log(1+e^{2T\tilde{C}})]T^2C}{\beta}$.

682 Next, we bound the first and second terms in (23) respectively. We start with the first term:

$$\begin{aligned}
& E \left[\left\| E[g_{\text{ce}}(\theta_k) - \nabla L_{\text{ce}}(\theta_k)] \right\| \right] \\
& = \left\| E_{x \sim \mathcal{D}, y \sim \pi_k(\cdot|x)} \left[l_{\text{ce}}(\theta; (x, y)) \nabla_{\theta} \log \pi_{r_{\theta}}(y|x) + \nabla_{\theta} l_{\text{ce}}(\theta; (x, y)) \right] \right. \\
& \quad \left. - E_{x \sim \mathcal{D}, y \sim \pi_{r_{\theta_k}}(\cdot|x)} \left[l_{\text{ce}}(\theta; (x, y)) \nabla_{\theta} \log \pi_{r_{\theta}}(y|x) + \nabla_{\theta} l_{\text{ce}}(\theta; (x, y)) \right] \right\| \\
& \leq \left\| E_{x \sim \mathcal{D}, y \sim \pi_k(\cdot|x)} \left[l_{\text{ce}}(\theta; (x, y)) \nabla_{\theta} \log \pi_{r_{\theta}}(y|x) \right] \right. \\
& \quad \left. - E_{x \sim \mathcal{D}, y \sim \pi_{r_{\theta_k}}(\cdot|x)} \left[l_{\text{ce}}(\theta; (x, y)) \nabla_{\theta} \log \pi_{r_{\theta}}(y|x) \right] \right\|
\end{aligned}$$

$$\begin{aligned}
& + \left\| E_{x \sim \mathcal{D}, y \sim \pi_k(\cdot|x)} [\nabla_\theta l_{ce}(\theta; (x, y))] - E_{x \sim \mathcal{D}, y \sim \pi_{r_{\theta_k}}(\cdot|x)} [\nabla_\theta l_{ce}(\theta; (x, y))] \right\| \\
& \stackrel{(f)}{\leq} \log(1 + e^{T\tilde{C}}) \left\| E_{x \sim \mathcal{D}, y \sim \pi_k(\cdot|x)} [\nabla_\theta \log \pi_{r_\theta}(y|x)] - E_{x \sim \mathcal{D}, y \sim \pi_{r_{\theta_k}}(\cdot|x)} [\nabla_\theta \log \pi_{r_\theta}(y|x)] \right\| \\
& + \left\| E_{x \sim \mathcal{D}, y \sim \pi_k(\cdot|x)} [\nabla_\theta l_{ce}(\theta; (x, y))] - E_{x \sim \mathcal{D}, y \sim \pi_{r_{\theta_k}}(\cdot|x)} [\nabla_\theta l_{ce}(\theta; (x, y))] \right\| \\
& \stackrel{(g)}{\leq} \log(1 + e^{T\tilde{C}}) \cdot 2TCC_d \sqrt{|\mathcal{S}| \times |\mathcal{A}|} E \left[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^* \|_\infty \right] \\
& + \left\| E_{x \sim \mathcal{D}, y \sim \pi_k(\cdot|x)} [\nabla_\theta l_{ce}(\theta; (x, y))] - E_{x \sim \mathcal{D}, y \sim \pi_{r_{\theta_k}}(\cdot|x)} [\nabla_\theta l_{ce}(\theta; (x, y))] \right\| \\
& \stackrel{(h)}{\leq} \log(1 + e^{T\tilde{C}}) \cdot 2TCC_d \sqrt{|\mathcal{S}| \times |\mathcal{A}|} E \left[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^* \|_\infty \right] \\
& + \left\| E_{x \sim \mathcal{D}, y \sim \pi_k(\cdot|x)} [\nabla_\theta r_\theta(x, y)] - E_{x \sim \mathcal{D}, y \sim \pi_{r_{\theta_k}}(\cdot|x)} [\nabla_\theta r_\theta(x, y)] \right\| \\
& \stackrel{(i)}{\leq} [\log(1 + e^{T\tilde{C}}) \cdot 2TCC_d + CC_d] \sqrt{|\mathcal{S}| \times |\mathcal{A}|} E \left[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^* \|_\infty \right], \tag{24}
\end{aligned}$$

683 where (f) follows from Lemma 5, (g) follows from (22), (h) follows from the expression (8) of
684 $\nabla_\theta l_{ce}$, and (i) follows from Claim 5.

685 Next, we bound the second term in (23):

$$\begin{aligned}
& E \left[\left\| E[g_{\text{dpo}}(\theta_k) - \nabla L_{\text{dpo}}(\theta_k)] \right\| \right] \\
& = \left\| E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_{r_{\theta_k}}(\cdot, \cdot|x)} [l_{\text{dpo}}(\theta; (x, y^+, y^-)) \nabla_\theta \log \pi_{r_\theta}(y^+, y^-|x) + \nabla_\theta l_{\text{dpo}}(\theta; (x, y^+, y^-))] \right. \\
& \quad \left. - E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_k(\cdot, \cdot|x)} [l_{\text{dpo}}(\theta; (x, y^+, y^-)) \nabla_\theta \log \pi_{r_\theta}(y^+, y^-|x) + \nabla_\theta l_{\text{dpo}}(\theta; (x, y^+, y^-))] \right\| \\
& \leq \left\| E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_{r_{\theta_k}}(\cdot, \cdot|x)} [l_{\text{dpo}}(\theta; (x, y^+, y^-)) \nabla_\theta \log \pi_{r_\theta}(y^+, y^-|x)] \right. \\
& \quad \left. - E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_k(\cdot, \cdot|x)} [l_{\text{dpo}}(\theta; (x, y^+, y^-)) \nabla_\theta \log \pi_{r_\theta}(y^+, y^-|x)] \right\| \\
& + \left\| E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_{r_{\theta_k}}(\cdot, \cdot|x)} [\nabla_\theta l_{\text{dpo}}(\theta; (x, y^+, y^-))] \right. \\
& \quad \left. - E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_k(\cdot, \cdot|x)} [\nabla_\theta l_{\text{dpo}}(\theta; (x, y^+, y^-))] \right\| \\
& \stackrel{(j)}{\leq} \log(1 + e^{2T\tilde{C}}) \left\| E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_{r_{\theta_k}}(\cdot, \cdot|x)} [\nabla_\theta \log \pi_{r_\theta}(y^+, y^-|x)] \right. \\
& \quad \left. - E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_k(\cdot, \cdot|x)} [\nabla_\theta \log \pi_{r_\theta}(y^+, y^-|x)] \right\| \\
& + \left\| E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_{r_{\theta_k}}(\cdot, \cdot|x)} [\nabla_\theta l_{\text{dpo}}(\theta; (x, y^+, y^-))] \right. \\
& \quad \left. - E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_k(\cdot, \cdot|x)} [\nabla_\theta l_{\text{dpo}}(\theta; (x, y^+, y^-))] \right\| \\
& \stackrel{(k)}{\leq} \log(1 + e^{2T\tilde{C}}) \cdot 4TCC_d \sqrt{|\mathcal{S}| \times |\mathcal{A}|} E \left[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^* \|_\infty \right] \\
& + \left\| E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_{r_{\theta_k}}(\cdot, \cdot|x)} [\nabla_\theta l_{\text{dpo}}(\theta; (x, y^+, y^-))] \right. \\
& \quad \left. - E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_k(\cdot, \cdot|x)} [\nabla_\theta l_{\text{dpo}}(\theta; (x, y^+, y^-))] \right\| \\
& \stackrel{(l)}{\leq} \log(1 + e^{2T\tilde{C}}) \cdot 4TCC_d \sqrt{|\mathcal{S}| \times |\mathcal{A}|} E \left[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^* \|_\infty \right] \\
& + \left\| E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_k(\cdot|x)} [\nabla_\theta r_\theta(x, y^+) - \nabla_\theta r_\theta(x, y^-)] \right. \\
& \quad \left. - E_{x \sim \mathcal{D}, (y^+, y^-) \sim \pi_{r_{\theta_k}}(\cdot|x)} [\nabla_\theta r_\theta(x, y^+) - \nabla_\theta r_\theta(x, y^-)] \right\| \\
& \stackrel{(m)}{\leq} [\log(1 + e^{2T\tilde{C}}) \cdot 4TCC_d 2CC_d] \sqrt{|\mathcal{S}| \times |\mathcal{A}|} E \left[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^* \|_\infty \right], \tag{25}
\end{aligned}$$

where (j) follows from Lemma 6, (k) follows from (22) and the fact that $\log \pi(y^+, y^- | x) = \log \pi(y^+ | x) + \log \pi(y^- | x)$, (l) follows from the expression (12) of $\nabla_{\theta} l_{\text{dpo}}$, and (m) follows from Claim 5.

By plugging (24) and (25) into (23), we have that

$$E[\langle \nabla L(\theta_k), E[g(\theta_k) - \nabla L(\theta_k) | \theta_k] \rangle] \leq C_{\Delta} E[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_{\infty}], \quad (26)$$

where $C_{\Delta} = \frac{[\log(1 + e^{T\tilde{C}}) \cdot 2TCC_d + CC_d] \sqrt{|\mathcal{S}| \times |\mathcal{A}|} + \lambda[\log(1 + e^{2T\tilde{C}}) \cdot 4TCC_d 2CC_d] \sqrt{|\mathcal{S}| \times |\mathcal{A}|}}{2}$.

Recall from (21) that

$$\begin{aligned} E[L(\theta_{k+1})] &\leq E[L(\theta_k)] - \alpha E[\langle \nabla L(\theta_k), E[g(\theta_k) - \nabla L(\theta_k) | \theta_k] \rangle] - \alpha E[\|\nabla L(\theta_k)\|^2] + \frac{C_L C_g \alpha^2}{2} \\ &\stackrel{(n)}{\leq} E[L(\theta_k)] + \alpha C_L C_{\Delta} E[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_{\infty}] - \alpha E[\|\nabla L(\theta_k)\|^2] + \frac{C_L C_g \alpha^2}{2}, \end{aligned}$$

where (n) follows from Lemma 7 and (26). Therefore, we have that

$$\alpha E[\|\nabla L(\theta_k)\|^2] \leq E[L(\theta_k)] - E[L(\theta_{k+1})] + \alpha C_L C_{\Delta} E[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_{\infty}] + \frac{C_L C_g \alpha^2}{2}.$$

Telescoping from 0 to $K - 1$ on both sides, we can get that

$$\begin{aligned} &\alpha \sum_{k=0}^{K-1} E[\|\nabla L(\theta_k)\|^2] \\ &\leq E[L(\theta_0)] - E[L(\theta_K)] + \alpha C_L C_{\Delta} \sum_{k=0}^{K-1} E[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_{\infty}] + \sum_{k=0}^{K-1} \frac{C_L C_g \alpha^2}{2}. \end{aligned}$$

Dividing both sides by αK :

$$\begin{aligned} &\frac{1}{K} \sum_{k=0}^{K-1} E[\|\nabla L(\theta_k)\|^2] \\ &\leq \frac{1}{\alpha K} [E[L(\theta_0)] - E[L(\theta_K)]] + \frac{C_L C_{\Delta}}{K} \sum_{k=0}^{K-1} E[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_{\infty}] + \frac{C_L C_g \alpha}{2} \\ &\stackrel{(o)}{\leq} \frac{1}{\alpha K} [E[L(\theta_0)] - E[L(\theta_K)]] \\ &\quad + \frac{2\hat{C} C_L C_{\Delta}}{1 - \gamma} \left[(1 + 2\lambda)TC + \frac{[2\log(1 + e^{T\tilde{C}}) + 4\lambda\log(1 + e^{2T\tilde{C}})]T^2 C}{\beta} \right] \cdot \frac{1}{K} \sum_{k=1}^K \alpha \\ &\quad + \frac{\gamma C_L C_{\Delta}}{(1 - \gamma)K} (\|Q_{r_{\theta_0}}^* - Q_{r_{\theta_0}}^{\pi_0}\|_{\infty} - \|Q_{r_{\theta_K}}^* - Q_{r_{\theta_K}}^{\pi_K}\|_{\infty}) + \frac{C_L C_{\Delta}}{K} \|Q_{r_{\theta_0}}^{\pi_0} - Q_{r_{\theta_0}}^*\|_{\infty} + \frac{C_L C_g \alpha}{2} \\ &= O\left(\frac{1}{\sqrt{K}}\right) + O\left(\frac{1}{K}\right), \quad (27) \end{aligned}$$

where (o) follows from (20).

Note that $\nabla L(\theta_k) = \nabla L(\theta_k, \pi_{r_{\theta_k}})$ and we need to upper bound $\|\nabla L(\theta_k, \pi_k)\|$. Therefore, we need to bound the difference:

$$\begin{aligned} &E[\|\nabla L(\theta_k, \pi_{r_{\theta_k}}) - \nabla L(\theta_k)\|^2] \\ &\leq E[2\|\nabla L_{\text{ce}}(\theta_k, \pi_{r_{\theta_k}}) - \nabla L_{\text{ce}}(\theta_k)\|^2 + 2\lambda^2 \|\nabla L_{\text{dpo}}(\theta_k, \pi_{r_{\theta_k}}) - \nabla L_{\text{dpo}}(\theta_k)\|^2] \\ &\stackrel{(p)}{\leq} C_1 E[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_{\infty}^2], \quad (28) \end{aligned}$$

where (p) follows from (24) and (25) such that $C_1 = 2 \left[\left(\log(1 + e^{T\tilde{C}}) \cdot 2TCC_d + CC_d \right)^2 + \lambda^2 \left(\log(1 + e^{2T\tilde{C}}) \cdot 4TCC_d 2CC_d \right)^2 \right] (|\mathcal{S}| \times |\mathcal{A}|)$.

701 Telescoping $E[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_\infty^2]$ from 0 to $K-1$, we can get that

$$\begin{aligned}
& \sum_{k=0}^{K-1} E[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_\infty^2] = E[\|Q_{r_{\theta_0}}^{\pi_0} - Q_{r_{\theta_0}}^*\|_\infty^2] + \sum_{k=1}^{K-1} E[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_\infty^2] \\
& \stackrel{(q)}{\leq} E[\|Q_{r_{\theta_0}}^{\pi_0} - Q_{r_{\theta_0}}^*\|_\infty^2] + \sum_{k=1}^{K-1} 8\hat{C}^2 E[\|\theta_k - \theta_{k-1}\|^2] + 2\gamma^2 E[\|Q_{r_{\theta_{k-1}}}^* - Q_{r_{\theta_{k-1}}}^{\pi_{k-1}}\|_\infty^2] \\
& = E[\|Q_{r_{\theta_0}}^{\pi_0} - Q_{r_{\theta_0}}^*\|_\infty^2] + \sum_{k=1}^{K-1} 8\hat{C}^2 \alpha^2 E[\|g(\theta_{k-1})\|^2] + 2\gamma^2 E[\|Q_{r_{\theta_{k-1}}}^* - Q_{r_{\theta_{k-1}}}^{\pi_{k-1}}\|_\infty^2] \\
& \stackrel{(r)}{\leq} \sum_{k=1}^{K-1} 8\hat{C}^2 \alpha^2 \left[(1+2\lambda)TC + \frac{[2\log(1+e^{T\tilde{C}}) + 4\lambda\log(1+e^{2T\tilde{C}})]T^2C}{\beta} \right]^2 \\
& \quad + 2\gamma^2 E[\|Q_{r_{\theta_{K-1}}}^* - Q_{r_{\theta_{K-1}}}^{\pi_{K-1}}\|_\infty^2] + E[\|Q_{r_{\theta_0}}^{\pi_0} - Q_{r_{\theta_0}}^*\|_\infty^2],
\end{aligned}$$

702 where (q) follows from (19) and that $(a+b)^2 \leq 2(a^2 + b^2)$, and (r) follows from Lemma 7.

703 Subtracting both sides by $2\sum_{k=0}^{K-1} \gamma^2 E[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_\infty^2]$, we can get that

$$\begin{aligned}
& (1-2\gamma^2) \sum_{k=0}^{K-1} E[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_\infty^2] \\
& \leq E[\|Q_{r_{\theta_0}}^* - Q_{r_{\theta_0}}^{\pi_0}\|_\infty^2] - 2\gamma^2 E[\|Q_{r_{\theta_{K-1}}}^* - Q_{r_{\theta_{K-1}}}^{\pi_{K-1}}\|_\infty^2] \\
& \quad + \sum_{k=1}^{K-1} 8\hat{C}^2 \alpha^2 \left[(1+2\lambda)TC + \frac{[2\log(1+e^{T\tilde{C}}) + 4\lambda\log(1+e^{2T\tilde{C}})]T^2C}{\beta} \right]^2. \tag{29}
\end{aligned}$$

704 Therefore, we can see that

$$\begin{aligned}
& \frac{1}{K} \sum_{k=0}^{K-1} E[\|\nabla L(\theta_k, \pi_{r_{\theta_k}}) - \nabla L(\theta_k)\|^2] \\
& \stackrel{(s)}{\leq} \frac{1}{K} \sum_{k=0}^{K-1} C_1 E[\|Q_{r_{\theta_k}}^{\pi_k} - Q_{r_{\theta_k}}^*\|_\infty^2] \\
& \stackrel{(t)}{\leq} \frac{C_1}{(1-2\gamma^2)K} \left\{ E[\|Q_{r_{\theta_0}}^* - Q_{r_{\theta_0}}^{\pi_0}\|_\infty^2] - 2\gamma^2 E[\|Q_{r_{\theta_{K-1}}}^* - Q_{r_{\theta_{K-1}}}^{\pi_{K-1}}\|_\infty^2] \right. \\
& \quad \left. + \sum_{k=1}^{K-1} 8\hat{C}^2 \alpha^2 \left[(1+2\lambda)TC + \frac{[2\log(1+e^{T\tilde{C}}) + 4\lambda\log(1+e^{2T\tilde{C}})]T^2C}{\beta} \right]^2 \right\} \\
& = O\left(\frac{1}{K}\right), \tag{30}
\end{aligned}$$

705 where (s) follows from (28) and (t) follows from (29).

706 By combining (27) and (30), we can reach that

$$\begin{aligned}
& \frac{1}{K} \sum_{k=0}^{K-1} E[\|\nabla L(\theta_k, \pi_{r_{\theta_k}})\|^2] \leq \frac{2}{K} \sum_{k=0}^{K-1} E[\|\nabla L(\theta_k, \pi_{r_{\theta_k}}) - \nabla L(\theta_k)\|^2] + \frac{2}{K} \sum_{k=0}^{K-1} E[\|\nabla L(\theta_k)\|^2] \\
& = O\left(\frac{1}{\sqrt{K}}\right) + O\left(\frac{1}{K}\right).
\end{aligned}$$

707 C Experiment Details

708 The code was run on eight A100 80G GPUs. The choices of the hyperparameters are the same across
709 benchmarks and models. The number of rollouts for each prompt is $m = 8$, the KL coefficient

710 is $\beta = 0.005$, the discount factor $\gamma = 1$, the weighting coefficient is 0.1, the number of training
711 iterations is $K = 450$, and the batch size of each iteration is $B = 512$. Both the reward model and
712 policy model use AdamW as the optimizer and the learning rate is 10^{-6} . The maximum prompt
713 length is 4096. For each turn, the maximum response length is 500. When the RL optimizer is PPO,
714 the critic uses AdamW as the optimizer with a learning rate of 10^{-5} .

715 C.1 Baselines

Prompt for LLM judge

You are an expert evaluator for multi-turn search-augmented reasoning systems. Given a user prompt, ground truth answer, and multi-turn generated response, evaluate each turn's effectiveness and compliance.

EVALUATION TASK

Assess each turn's format compliance, content quality, and contribution toward the ground truth answer.

SCORING CRITERIA

FINAL TURN (Last Turn) - Score Range: [-1.0 to 1.0]

Format Compliance:

- Required: '<think>...</think><answer>...</answer>' (tags only, once each, in order)
- Answer in '<answer>' tag must not exceed 5 tokens

Answer Correctness:

- Correct and complete answer in '<answer>' tag that matches the ground truth

Scoring Rules:

- If format is incorrect: Final Turn Score = -1.0
- If format is correct, answer is incorrect: Final Turn Score = 0.2
- If format is correct, answer is correct: Final Turn Score = 1.0

INTERMEDIATE TURNS - Score Range: [-1.0 to 1.0]

Format Compliance:

- Required: '<think>...</think><search>...</search><information>...</information>' (tags only, once each, in order)
- Correct format: +0.1
- Incorrect format: -0.2

Information Quality:

- Relevant information in '<information>' tag that helps toward the ground truth answer (e.g., ground truth exists in the retrieved result within '<information>' tag): +0.3
- Irrelevant or unhelpful information in '<information>' tag: +0.0

Search Efficiency Penalty:

- Number of searches = Total count of '<search>' tags across all turns from Turn 1 up to and including the current turn
- Search penalty = Number of searches $\times$ (-0.1)
- Encourages finding answers with fewer searches

****Intermediate Turn Score = Format Compliance + Information Quality + Search Penalty****

OUTPUT FORMAT

Provide your evaluation using ONLY these XML tags:

```
<reasoning>
Systematically evaluate each turn: check format compliance, assess content quality, calculate scores with clear
explanations
</reasoning>
```

```
<score>
Turn1: X.X
Turn2: X.X
Turn3: X.X
...
</score>
```

REQUIREMENTS:

- Must provide exactly $\{\text{len}(\text{turns})\}$ scores (one per turn)
- Use decimal format (e.g., 0.5, -0.3, 1.0)
- Use only the specified XML tags, no additional text

EVALUATION DATA

```
{prompt_text}
{turns_text}
{ground_truth_text}
**TURNS TO EVALUATE: {len(turns)}**
```

Your Evaluation

716

717 **Memory Comparison.** The baselines can be categorized into three classes based on the source of
718 reward signals: (1) rule-based reward (OR), (2) LLM-judge reward (LLMJ), and (3) learned reward

models (PRIME and iStar), alongside our method BARS. These categories incur different memory overhead due to their distinct reward mechanisms.

OR is the most memory-efficient method, as it computes a simple rule-based binary “exact match” signal without requiring additional models. In contrast, LLMJ is the most memory-intensive approach, as it relies on a large language model (GPT-OSS-120B) to generate reward signals during training.

Methods based on learned reward models, including PRIME, iStar, and BARS, have comparable memory overhead. Specifically, these methods require maintaining both a language model π_θ and a reference language model π_{ref} , which are typically of the same size as the underlying policy π_{r_θ} . Note that the reward model reuses the π_{ref} used for optimizing the policy model π_{r_θ} (see the lower-level RL problem (4)) so that the learned reward models require memory for an additional language model π_θ . In our experiments, we use Qwen2.5-3B and Llama3.1-8B, which are significantly smaller than the 120B-scale model used in LLMJ, resulting in substantially lower memory requirements.

The prompt for the turn-level LLM judge is provided in [9]. For the sake of completeness, we also provide it here.

C.2 Training Dynamics

In this section, we include the training and test curves on Qwen2.5-3B and the NQ-HotpotQA benchmark.

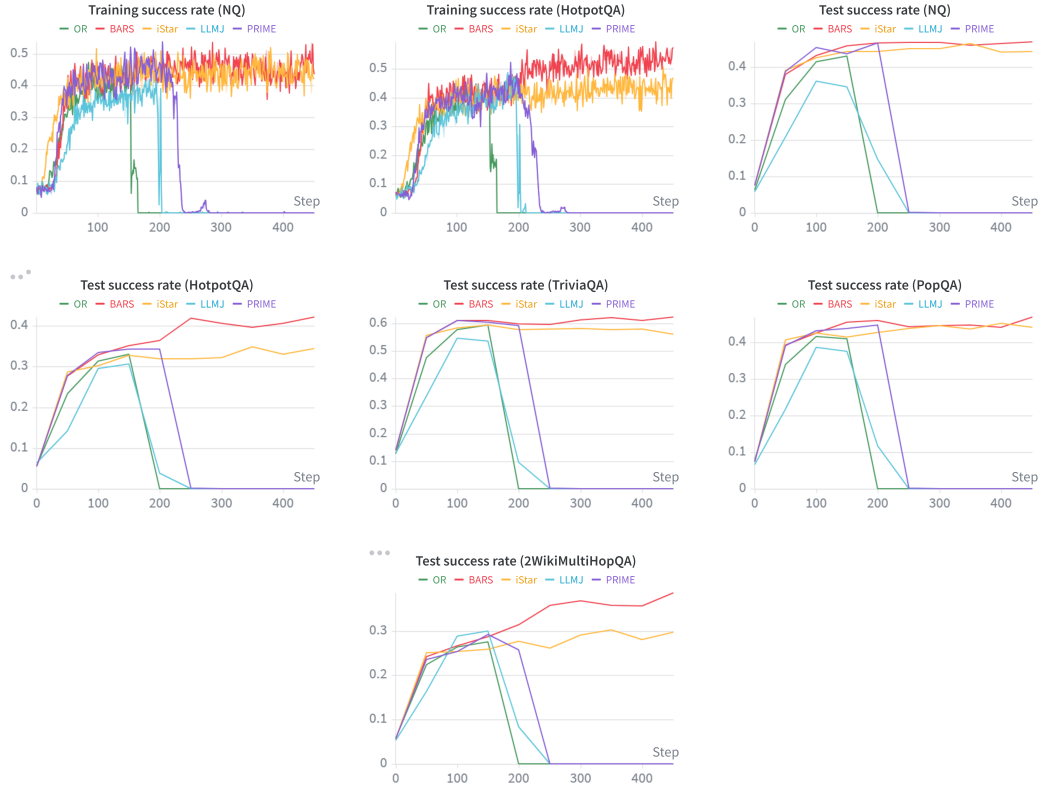


Figure 4: Training and test curves on Qwen2.5-3B and the NQ-HotpotQA benchmark.

Figure 4 shows the training and test success rates across multiple benchmarks. The training set consists of NQ and HotpotQA, where HotpotQA includes multi-hop questions, while the test set covers a broader collection of datasets including NQ, HotpotQA, TriviaQA, PopQA, and 2WikiMultiHopQA. BARS exhibits consistently stable training dynamics, with smooth and monotonic improvements that quickly plateau at a high performance level. In contrast, alternative methods such as OR, LLMJ, and PRIME demonstrate significant instability, including abrupt performance drops and even training collapse. This instability is also reflected in test performance, where these methods often degrade

743 sharply. Although iStar achieves stable training, it converges to a worse performance compared to
744 BARS. Overall, the results demonstrate that BARS not only achieves superior final accuracy but also
745 provides substantially improved training stability and strong generalization to both mixed single-hop
746 (NQ, TriviaQA, PopQA) and multi-hop evaluation benchmarks (HotpotQA, 2WikiMultiHopQA).

747 **D Limitations**

748 BARS depends on the quality of the outcome reward signal used in the upper-level objective. In our
749 experiments, we focus on binary success rewards, which are reliable since the ground truth answers
750 are given. In more complex real-world settings, outcome rewards may be noisy, delayed, or partially
751 specified, which could affect the quality of the learned shaping reward.

752 **E Potential Societal Impact**

753 This work contributes to improving reinforcement learning for agentic AI systems by addressing
754 sparse supervision and unstable optimization in multi-turn decision-making. By enabling more stable
755 and efficient learning from outcome-based signals, BARS can improve the reliability of AI agents in
756 applications such as information retrieval, decision support, and tool-augmented reasoning. More
757 broadly, our framework aligns with efforts in human-centered AI by improving credit assignment
758 while preserving the original task objective, which may help reduce unintended behaviors introduced
759 by heuristic or biased reward design. However, the ability to learn dense shaping rewards also
760 introduces potential risks. If the outcome reward is misaligned with human values or imperfectly
761 specified, the learned shaping reward may amplify these biases and lead to undesirable behaviors.

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